Agent-based summary

21/10-2025

Two competing events: (Q and decay x/τ)

$$\frac{dx}{dt} = Q - \frac{x}{\tau}$$

- Select time step for increase in x, $\Delta t_Q = -ln(ran_1)/Q$ by choosing a random number $ran_1 \in [0, 1]$.
- Select time step for decrease in x, $\Delta t_x = -ln(ran_2)\tau/x$ by choosing a random number $ran_2 \in [0, 1]$.
- If $\Delta t_Q < \Delta t_x$ then update $x = x + 1$ and set time counter $t = t + \Delta t_Q$.
- If $\Delta t_Q > \Delta t_x$ then update $x = x - 1$ and set time counter $t = t + \Delta t_x$.

SIR models:

$$\frac{dS}{dt} = -\lambda \cdot S \cdot I$$
$$\frac{dI}{dt} = \lambda \cdot S \cdot I - \gamma I$$
$$\frac{dR}{dt} = \gamma \cdot I$$

$$R_0 = \frac{\lambda}{\gamma},$$

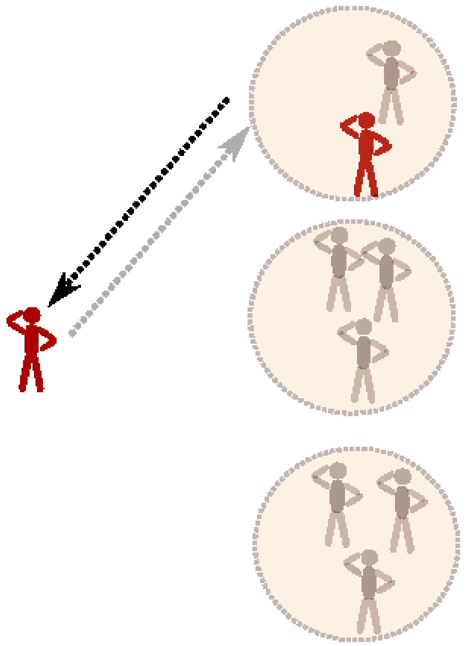
And infecting R0 people takes 1/ϒ time

Growth rate= dI/dt= (λ-ϒ) I (For S~1)

$$\frac{dln(S)}{dR} = -\lambda/\gamma = -R_0 \Rightarrow S(\infty) = e^{-R_0(1-S(\infty))}$$

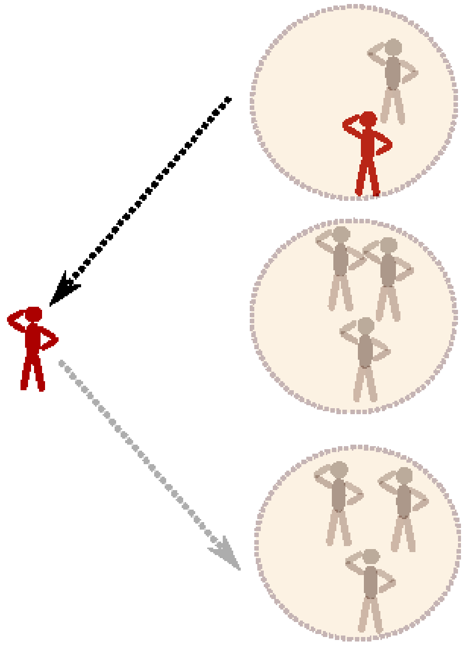
Agent based contra SIR=random mixing

Reality: Each man return
to same home every day



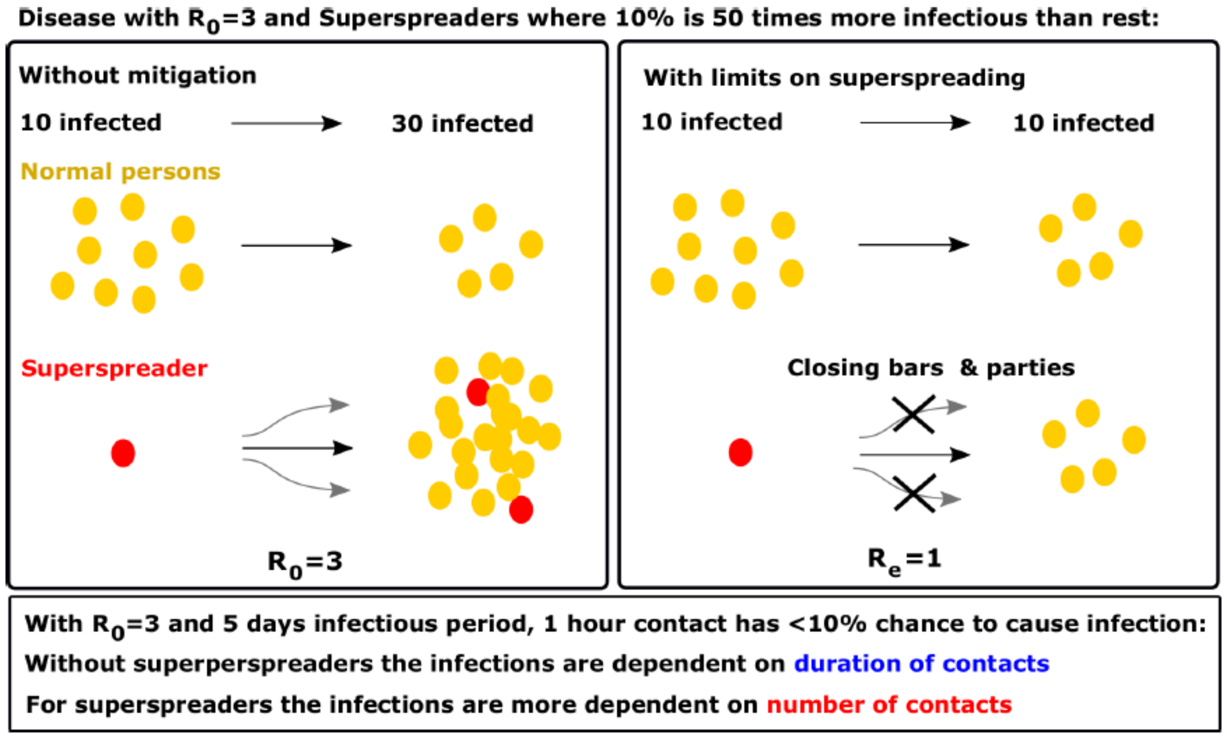
Here can only infect up to 2
new susceptibles

SIR simplification: Each man return
to a new home every day



Here he has unlimited infection
potential (one for each timestep)

- **An example where Agent-based models become crucial.**



Have to deal with confines of social Networks, and Finite time

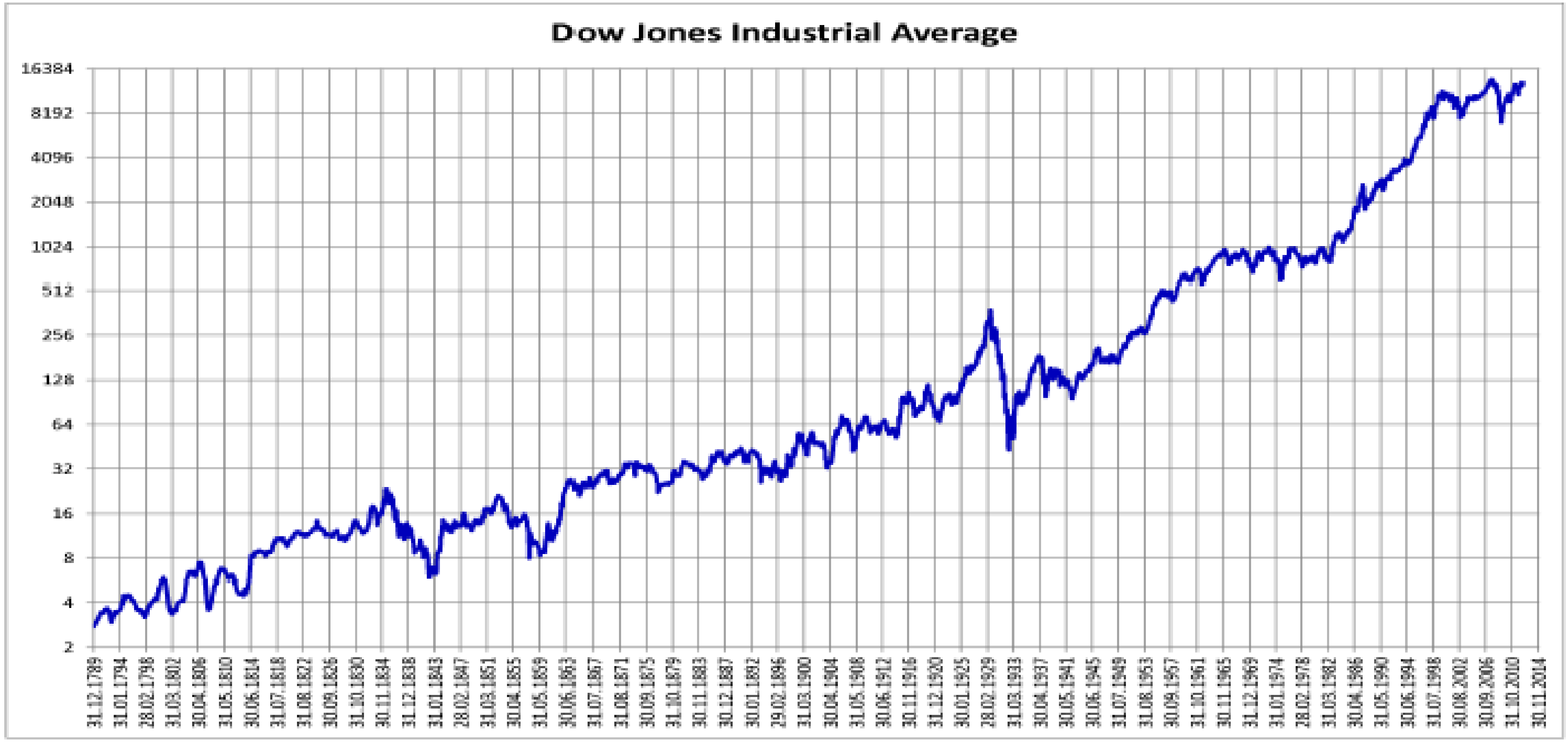
Econophysics

Dynamics of value / value of value

For then, since gold was soft and blunted easily, man would deem
it useless, but bronze was a metal held in high esteem.
Now the opposite: bronze is held cheap, while gold is prime.
And so the seasons of all things roll with the round of time:
What once was valuable, at length is held of no account,
while yet the worth of which was despised begin to mount.

Lucretius, De Renum Natura, Book 5 (60 years before crist)

Share Index over 230 years = average company value



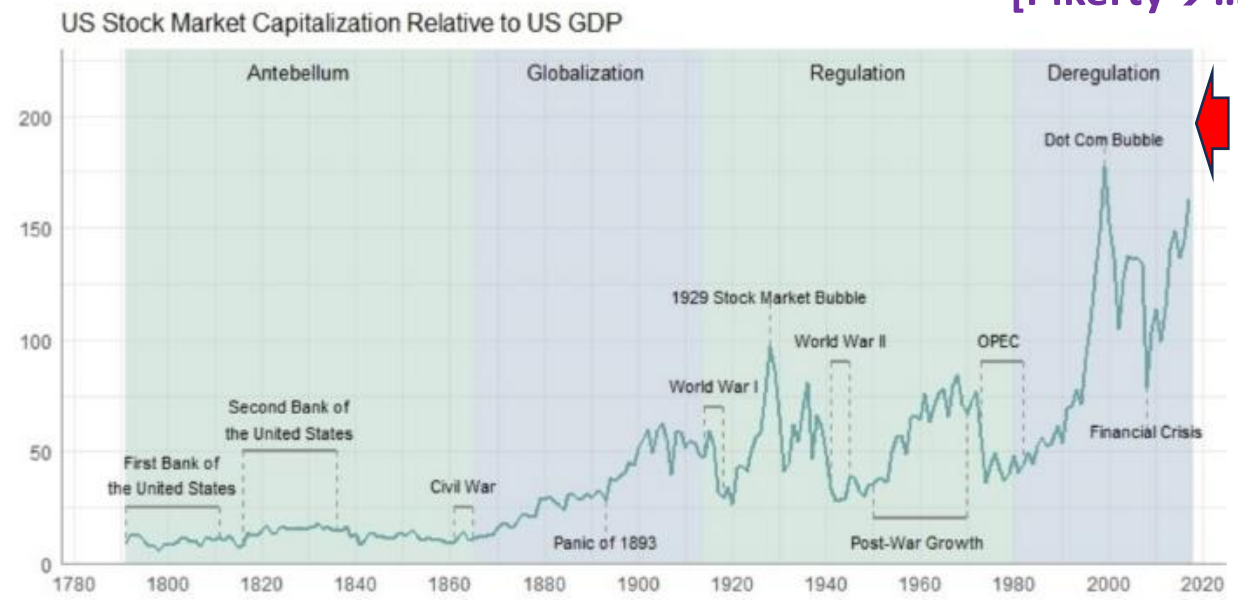
National dept: $\sim 10^8 \$$ *in the period 1800-1850 to* $\sim 5 \times 10^{12} \$$ *in year 2000.* Dow=10000)

2023: 33.000 and dept 30Trilllions (dept persistently grown more than company values)

…and Values of all shares relative to 1 years GDP:

[Pikerty→…]

USA:

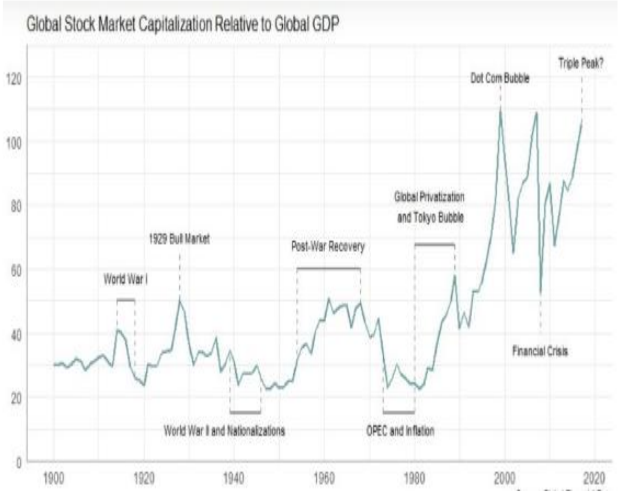


20.58

en.m.wikipedia.org

1	United States	44,719,661	194.5	4,266
2	China	13,214,311	83.0	4,154
3	Japan	6,718,220	122.2	3,754
4	Hong Kong	6,130,420	1,768.8	2,353
5	India	3,612,985[5]	103	5,270
6	France	2,823,000	84.9	457
7	United Kingdom	2,821,000	100	1,858
8	Canada	2,641,455	160.7	3,922
9	Saudi Arabia	2,429,102	347.0	207
10	Germany	2,284,109	60.0	438
11	South Korea	2,176,190	133.5	2,318
12	Taiwan	2,007,560	267.1	1,627
13	Switzerland	2,001,603	267.6	236
14	Australia	1,720,556	129.3	1,902
15	Netherlands	1,597,215[6]	132.3	103
16	Iran	1,218,392	635.5	367
17	South Africa	1,051,529	348.3	264

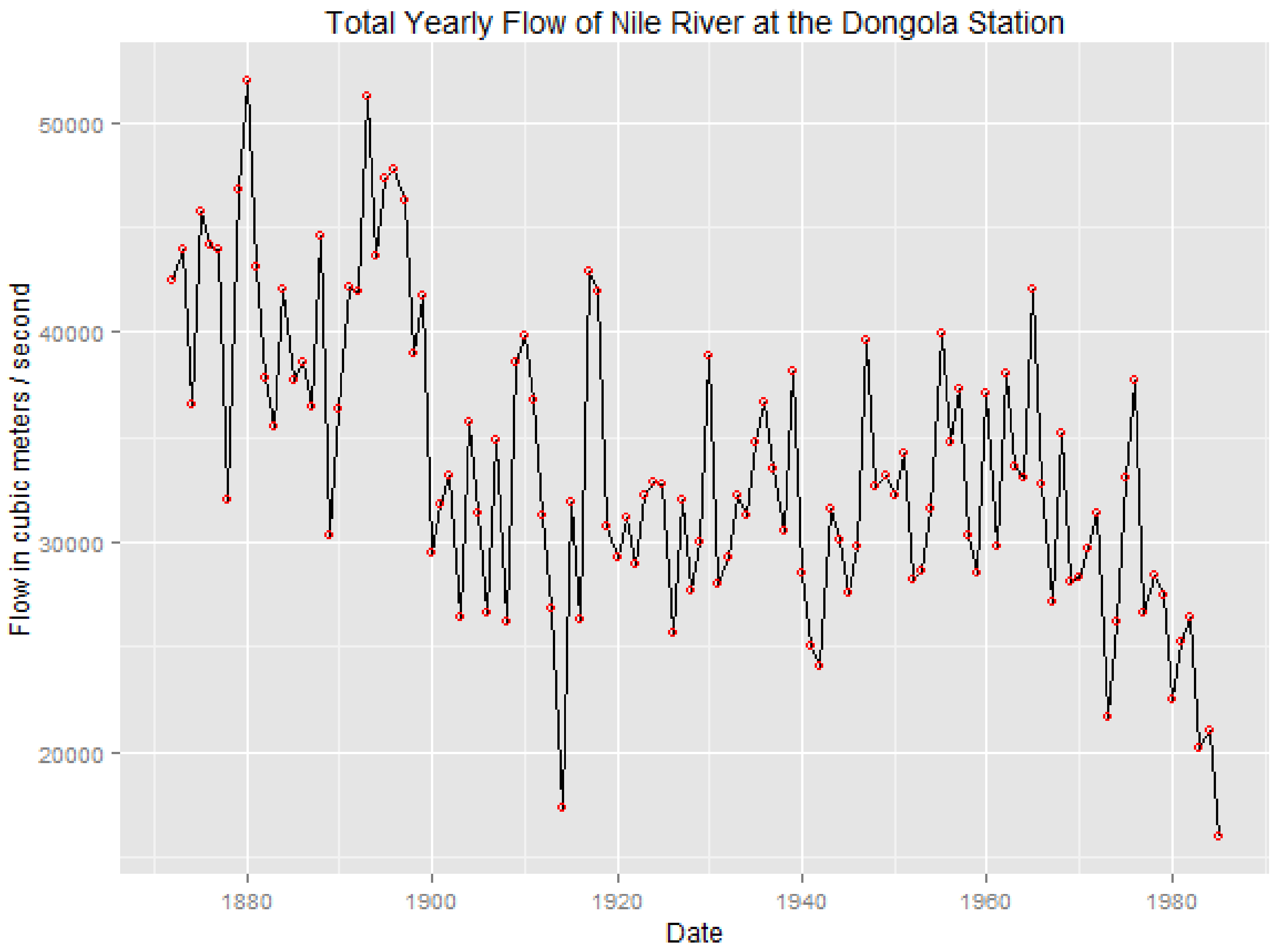
World:



US about 50% of the 92T$ world market Cap
But only about 26T$/86T$=30% of world GDP

Want to go beyond details….→
general characterization of timeseries

Other time series, e.g. the water flow in the Nile:



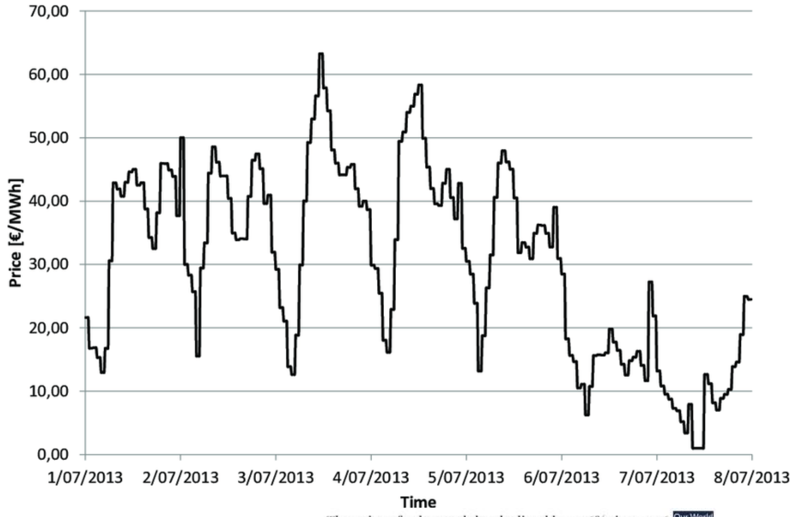
Studied by *Hurst* who introduced the **Hurst** exponent

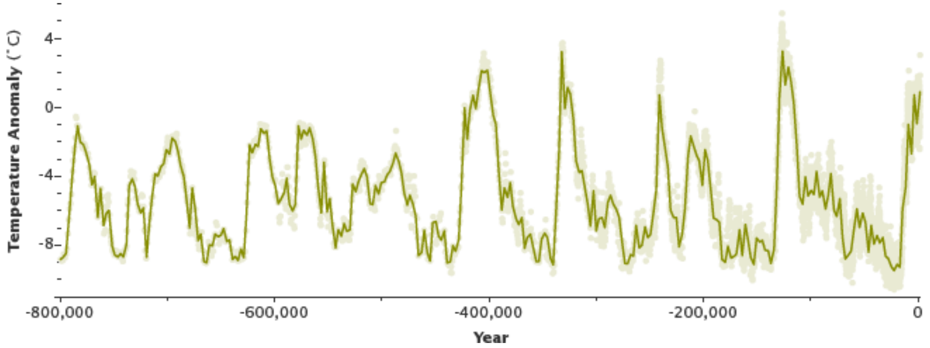
Some timeseries that matter:

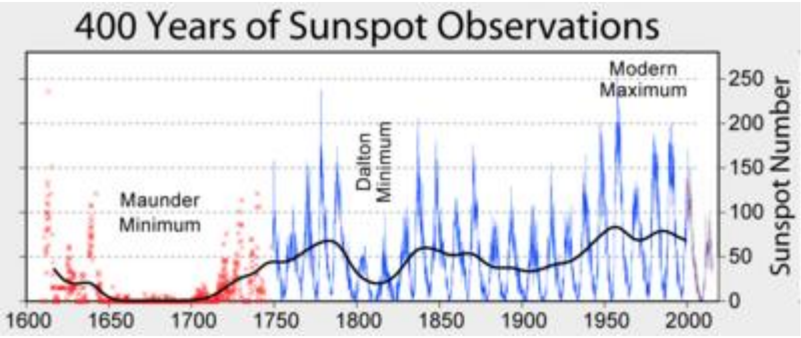
Crude Oil Price History Chart

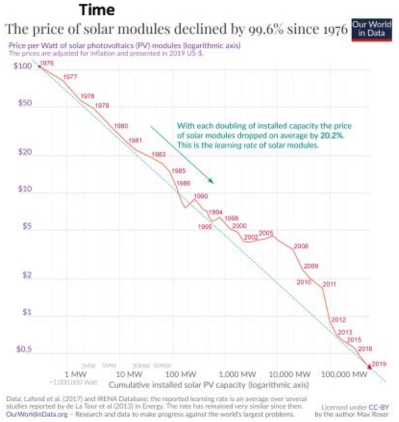
Natural gas price in the UK soars to highest level since 2005

UK wholesale gas price, pence per therm

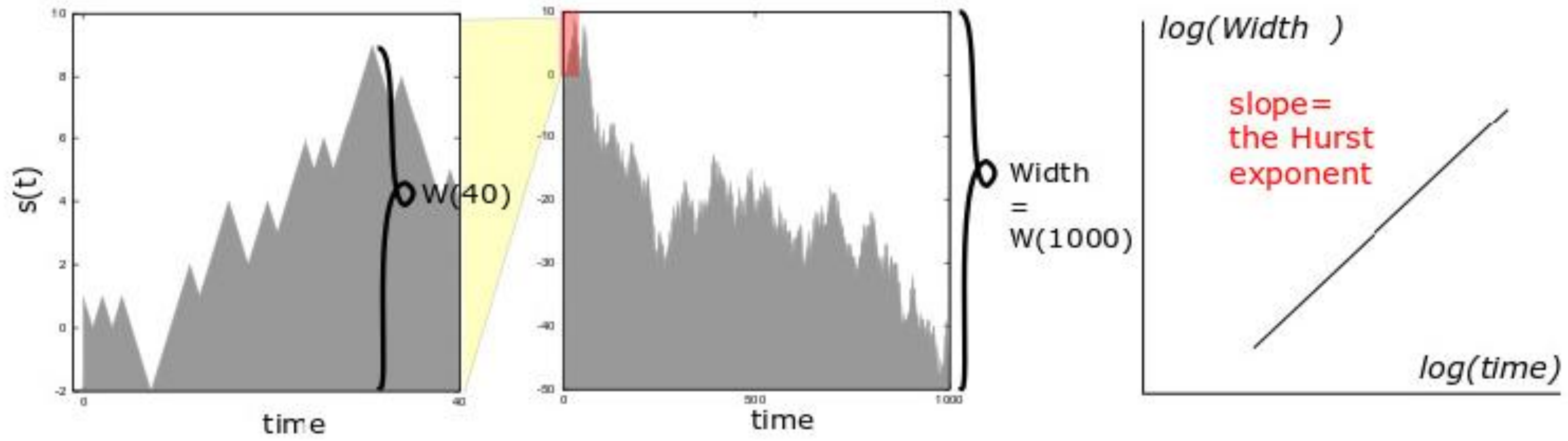






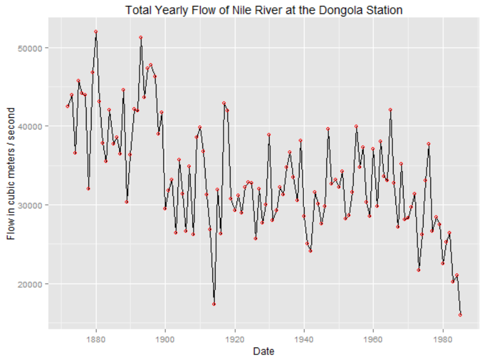


The “*Hurst*” exponent H and walks with D=2-H



$$\langle(\Delta s(T))^2\rangle = \langle(s(t+T) - s(t))^2\rangle_t \propto T^{2H}$$

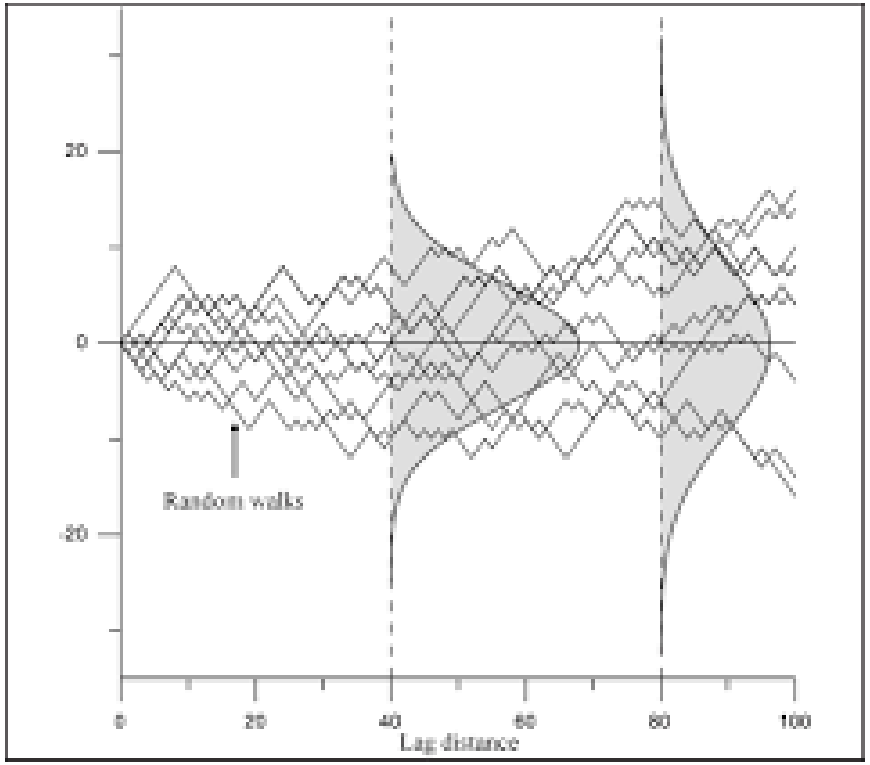
Invented by Hurst for studying water level in Nile:



Here H = 0.70→0.75

Dimension: D=2-H

Value of *H* for a random walk (of variable *s*):



$$\langle (\Delta s(T))^2 \rangle = \langle (s(t+T) - s(t))^2 \rangle_t \propto T^{2H}$$

Random walk: *H = 1/2*

Hurst exponent measured for *s=log(price)*

$$s(t) = \log(v(t))$$

Because price changes are relative (to scale where you are)

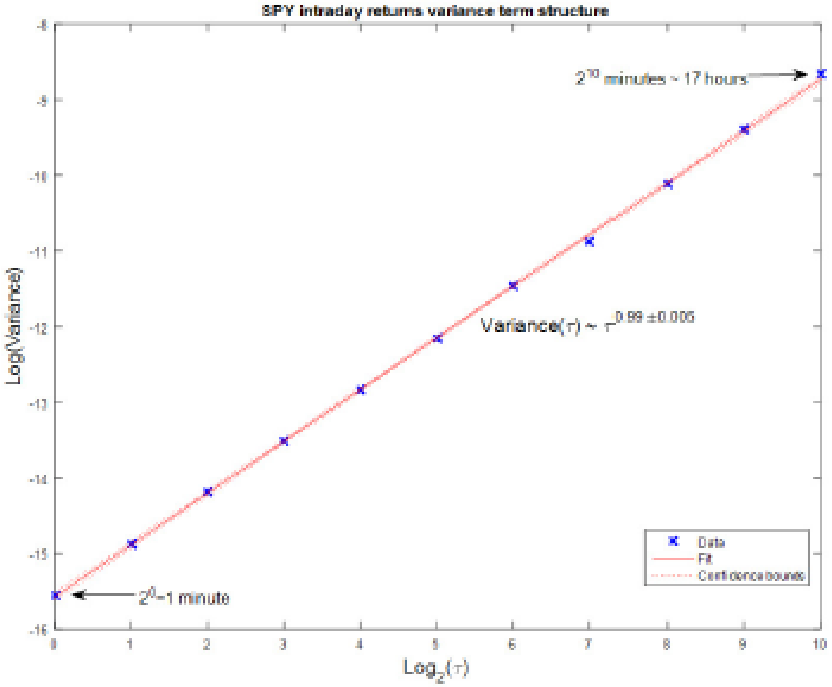
It is (Price(new)-Price(old))/Price(old) that matters

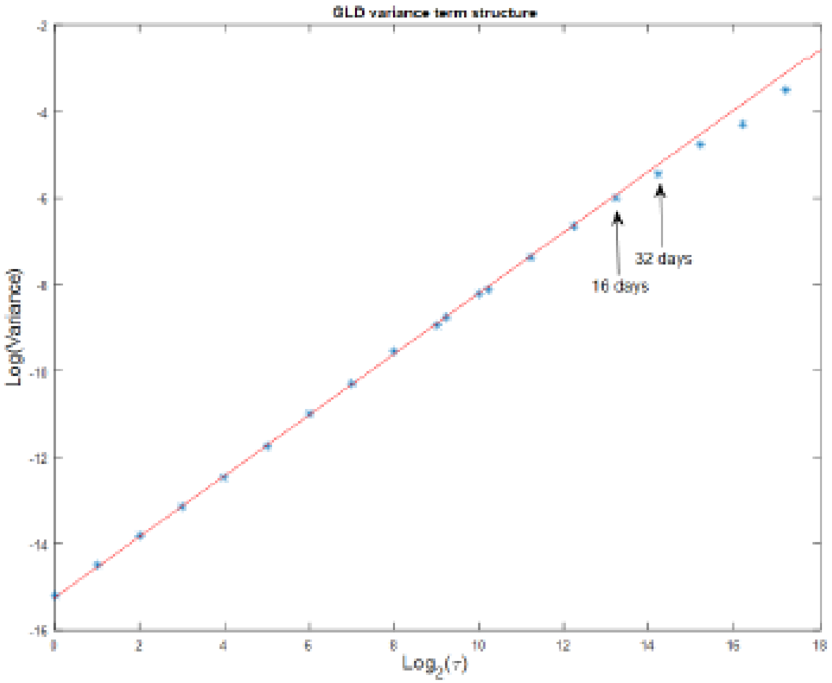
Not just the difference!!! (this is the amplification factor of investment)

Hurst exponent measured for *s=log(price)*

$$W^2(T) = \langle (\log v(t+T) - \log v(t))^2 \rangle_t = \langle (\Delta s(T))^2 \rangle \;;$$

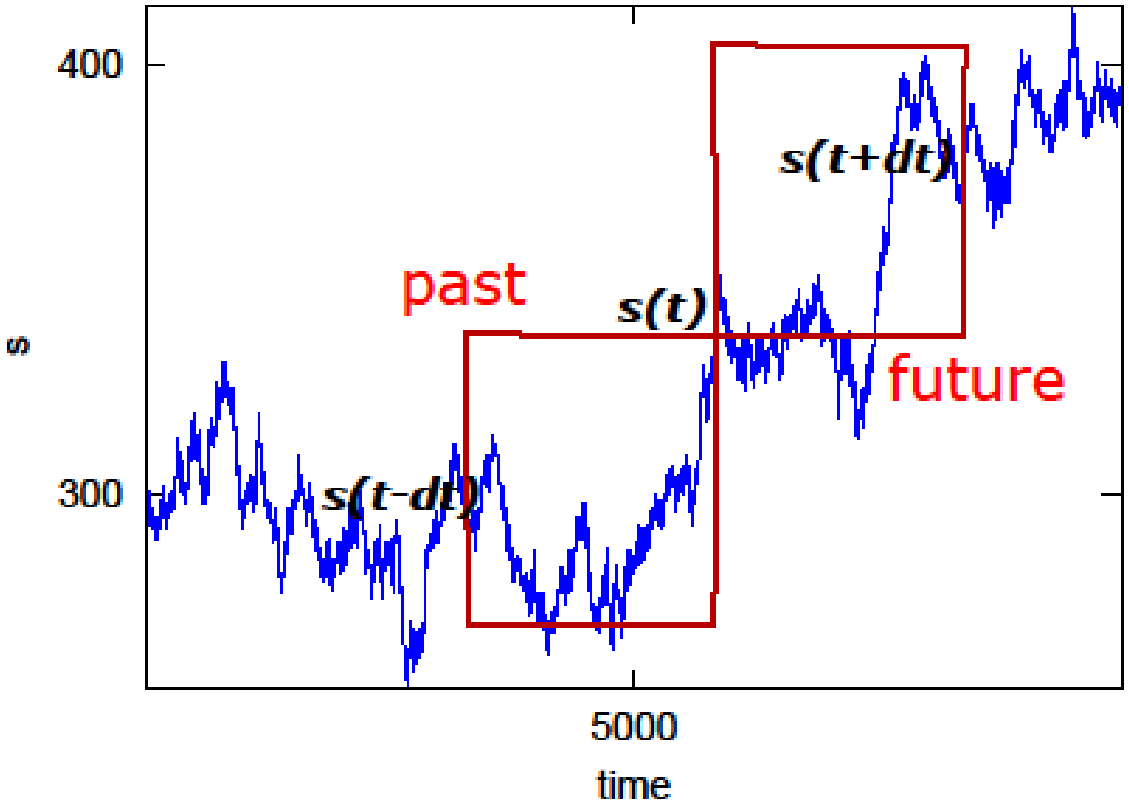
$$\langle (\Delta s(T))^2 \rangle = \langle (s(t+T) - s(t))^2 \rangle_t \propto T^{2H}$$

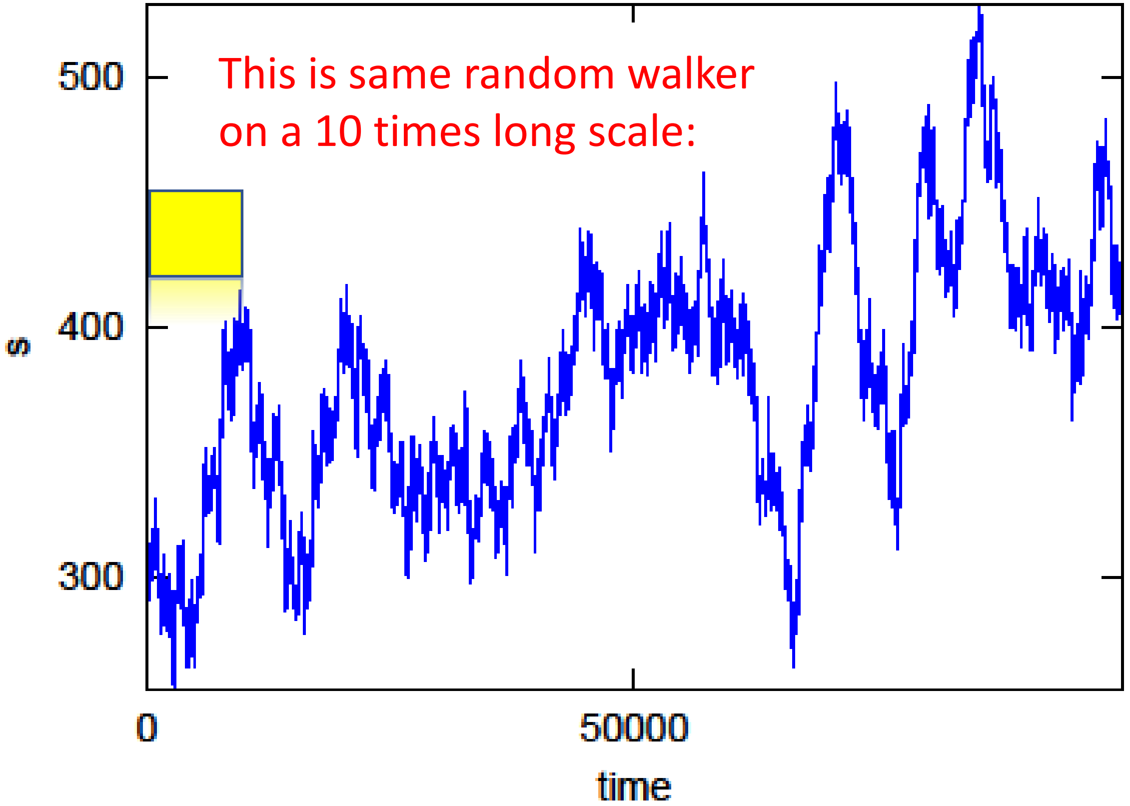




Correlation between past and future:

$$C \equiv \frac{\langle (s(t) - s(t-T)) \cdot (s(t+T) - s(t)) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t} = \frac{\langle -\Delta s(-T) \cdot \Delta s(T) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t}$$





$$\begin{aligned} f(T) &= \langle (\Delta s(T))^2 \rangle_x = \langle s^2(x+T) + s^2(x) - 2s(x)s(x+T) \rangle_x \\ &= 2(\langle s^2(x) \rangle - \langle s(x)s(x+T) \rangle_x) \propto T^{2H}, \end{aligned}$$

$$C \equiv \frac{\langle (s(t) - s(t-T)) \cdot (s(t+T) - s(t)) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t} = \frac{\langle -\Delta s(-T) \cdot \Delta s(T) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t}$$

$$\begin{aligned} \langle -\Delta s(-T) \Delta s(T) \rangle_x &= \langle -(s(x-T) - s(x))(s(x+T) - s(x)) \rangle_x \\ &= -\langle s(x-T) \cdot s(x+T) \rangle_x + \langle s(x-T) \cdot s(x) \rangle_x \\ &\quad + \langle s(x) \cdot s(x+T) \rangle_x - \langle s^2(x) \rangle_x \\ &= -\langle s(x) \cdot s(x+2T) \rangle_x + \langle s^2(x) \rangle_x \\ &\quad + 2\langle s(x) \cdot s(x+T) \rangle_x - 2\langle s^2(x) \rangle_x \\ &= \frac{1}{2} f(2T) - f(T). \end{aligned}$$

$$f(T) = const \cdot T^{2H} \Rightarrow C = \frac{\langle -\Delta s(-T) \cdot \Delta s(T) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t} = 2^{2H-1} - 1$$

Simple Translation, consider one time interval: ΔT

To get an interpretation of the above correlation, consider a stock that on a time scale T follow the trend with probability p and reverses it with probability $1-p$. The variance for one step of this walk is $\langle(\Delta s(T))^2\rangle_t = 1$. The numerator in eq. 5.5 is given by the sum of two contributions, one for following the trend,and one for reversing the trend

$$\langle -\Delta s(-T)\cdot\Delta s(T)\rangle_t = 1\cdot p\cdot 1 + 1\cdot(1-p)\cdot(-1) = 2p-1\,.$$

$$2p-1 = 2^{2H-1}-1 \text{ or } p = 4^{H-1}.$$

That means that the value of p to fit the top eq. with expected H exponent should be

$$\text{Propability (follow the trend)} \sim 4^{H-1}$$

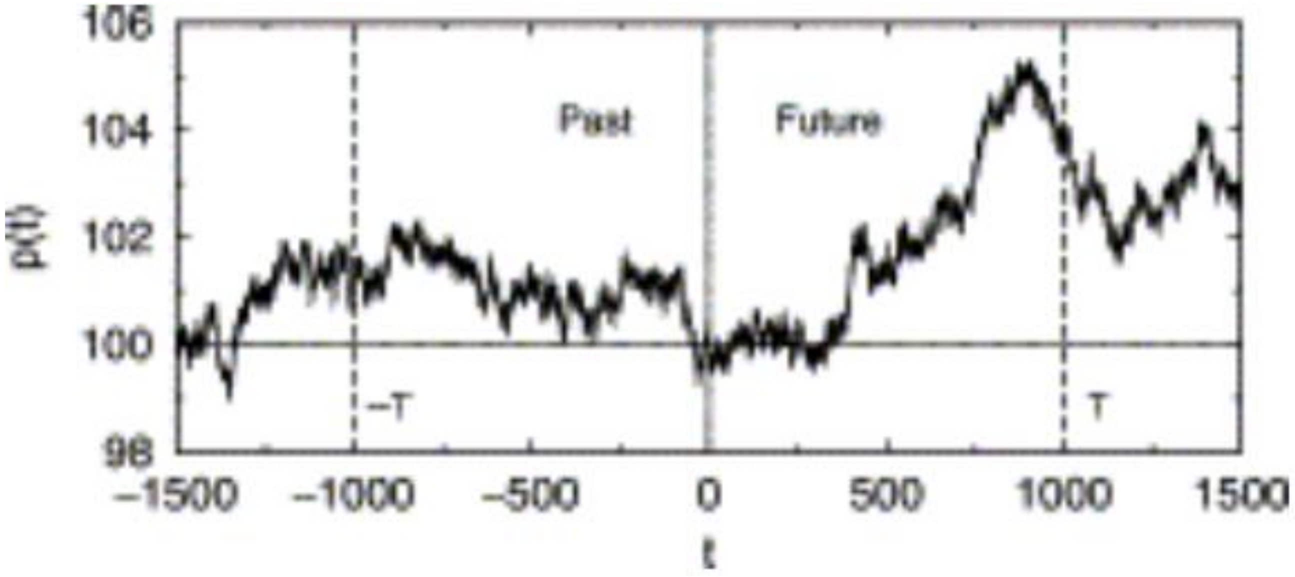
AND that THIS SHOULD BE TRUE ON ANY SCALE……. VERY DIFFICULT TO CONSTRUCT

Important:

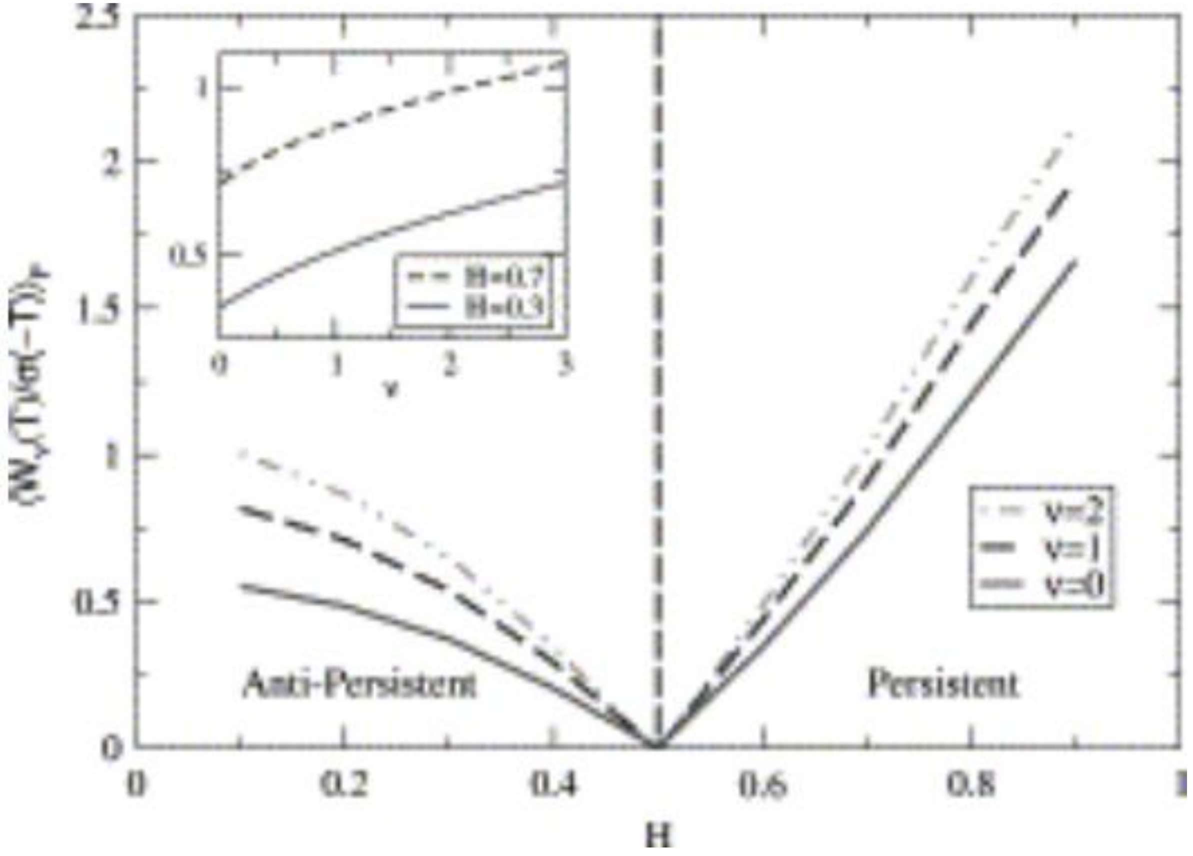
$$H>0.5 \;\underset{\not\Leftarrow}{\Rightarrow}\; p_{\text{follow trend}}=4^{H-1}>0.5$$

The back implication would require that it SHOULD be true on any scale

Bet according to trend: What is average profit

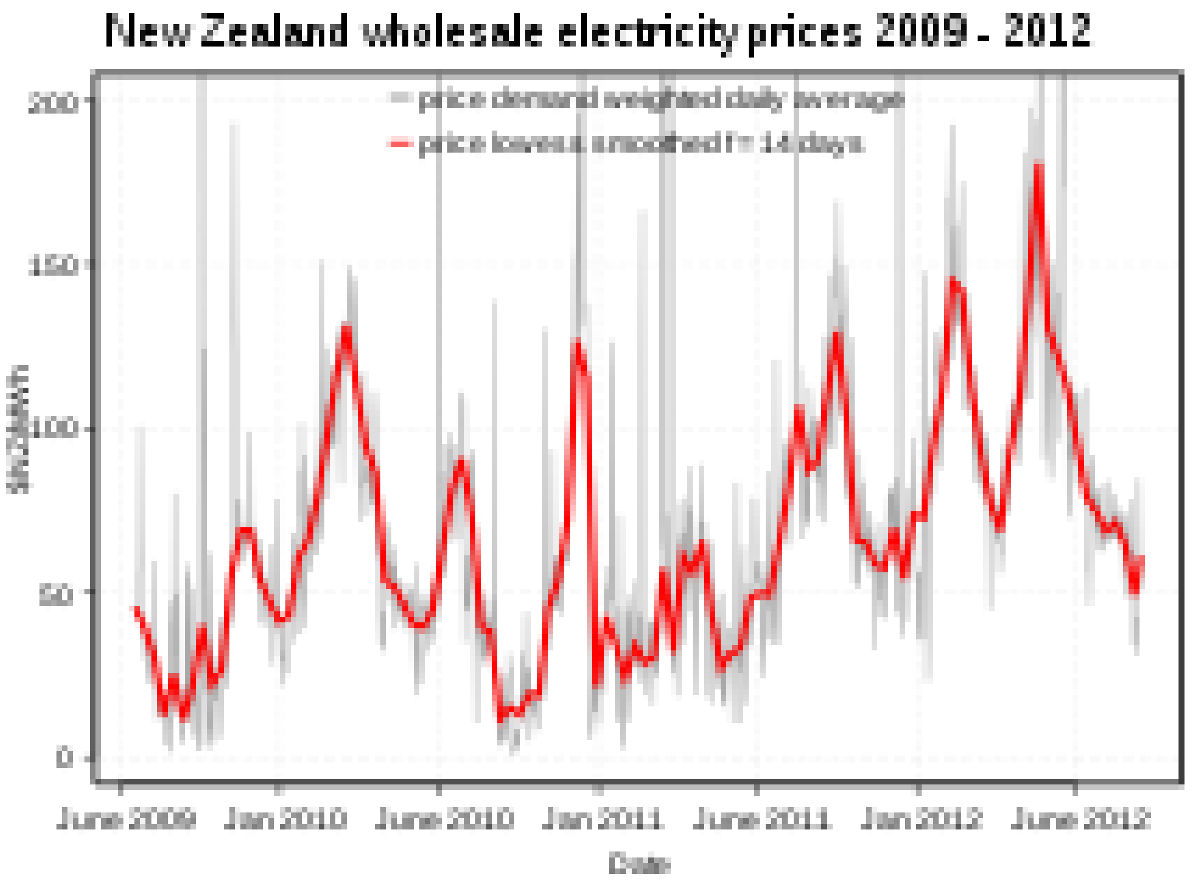


$$C = \frac{\langle -\Delta s(-T) \cdot \Delta s(T) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t} = 2^{2H-1} - 1$$



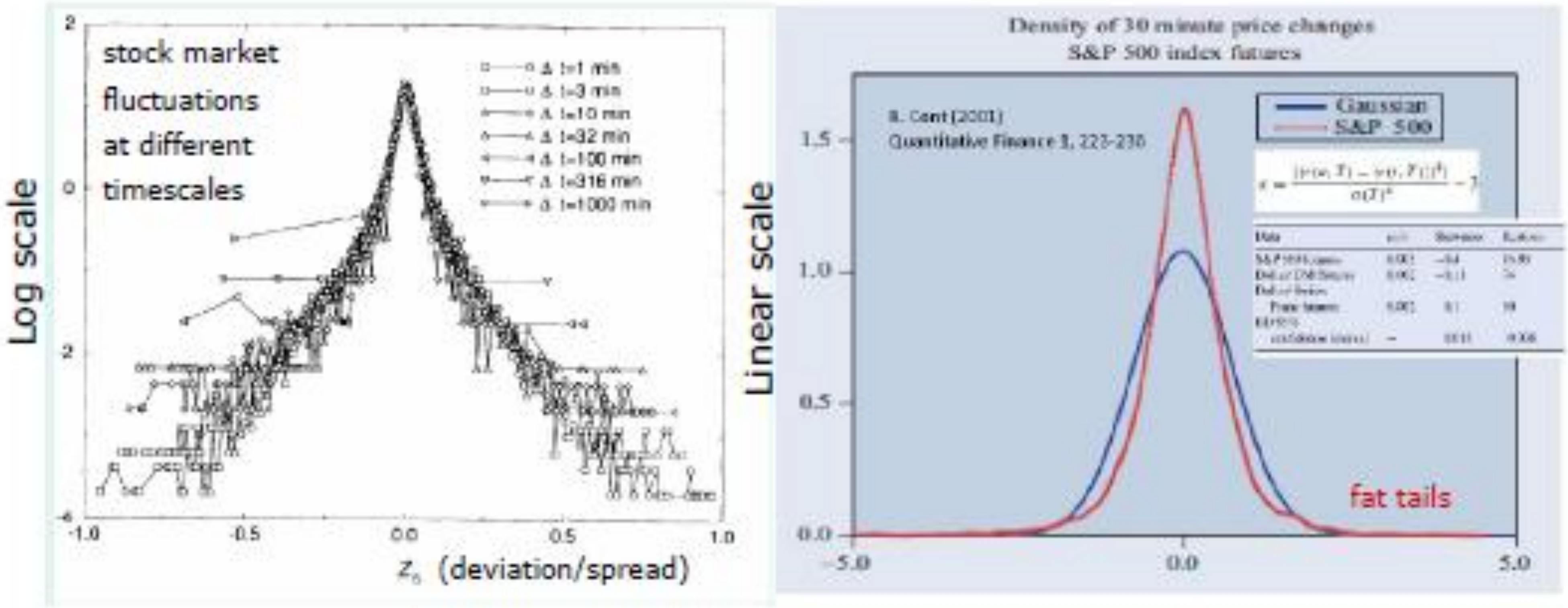
Profit in unit of spread of price after given timescale

Example:
Electricity market is anti-persistent:



(in fact *H=0.4*)

Stock market Deviations from random walk



Short terms fluctuations not Gaussian, 4'th moment not defined ($1/\Delta s^{4.5}$).

Indeed there exist large fluctuations:

Santa Ana Register

FINAL EDITION

BILLIONS LOST AS STOCKS CRASH

Pantages In Jail Awaits Sentence To State Prison

GUILTY SAYS JURY AFTER MANY HOURS

SENATE "FARM BLOC" STARTS BATTLE FOR HIGHER TARIFFS

JURY BELIEVES HER STORY

MAINSTAYS OF MARKET IN PLUNGE

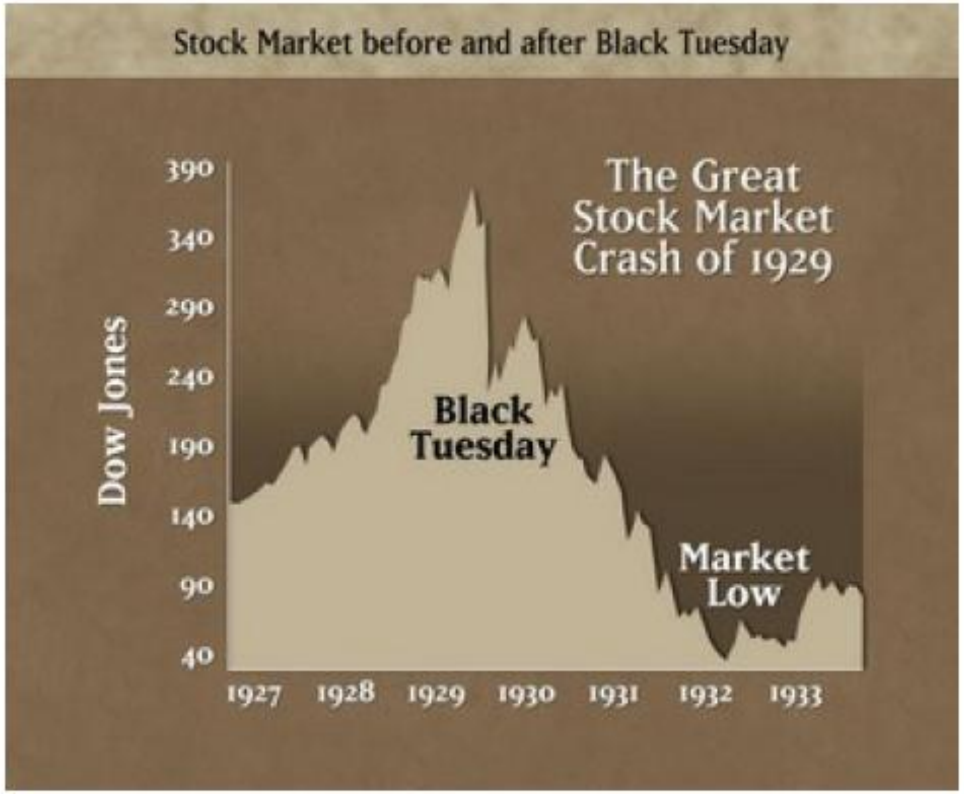
STATE PLANS TO ROUND UP TAX DODGERS

BINGHAM TAKES EXCEPTION TO SENATE LOBBY

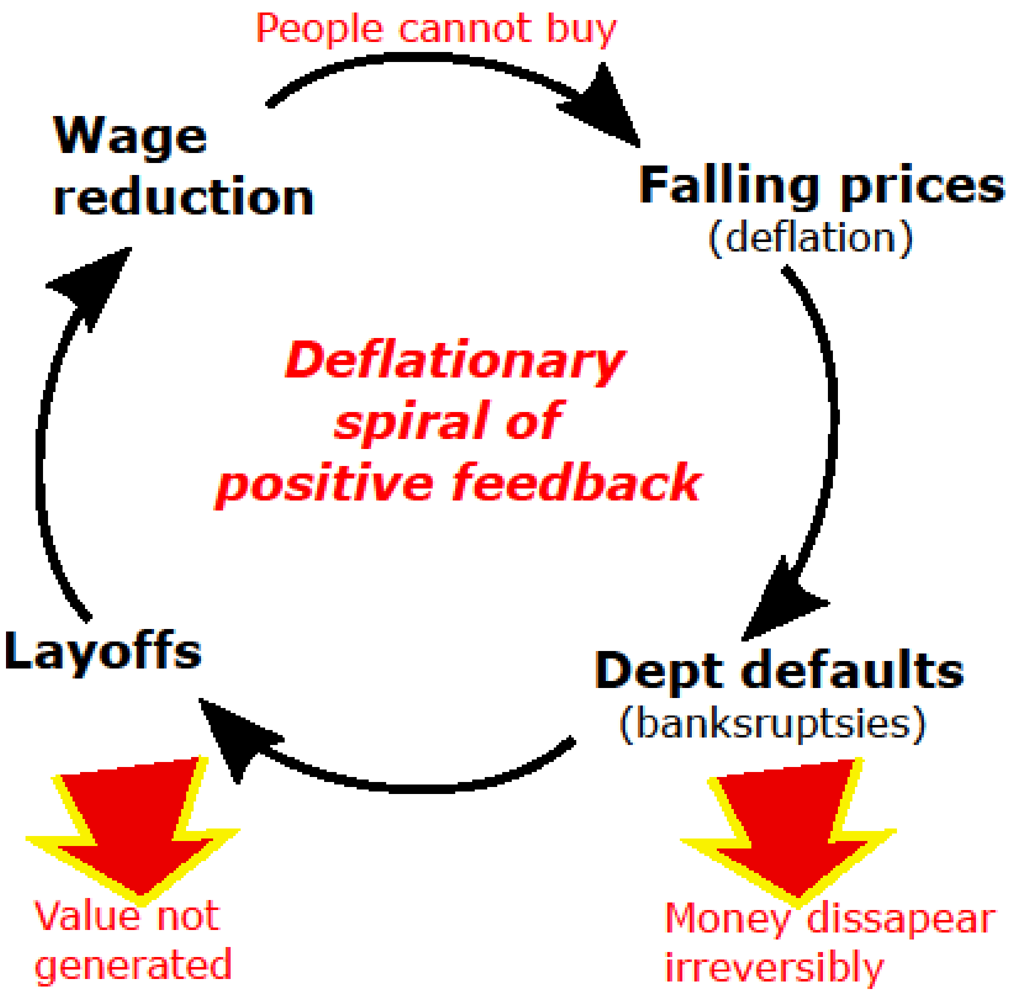
Alliance Of Two Factions May Win Out

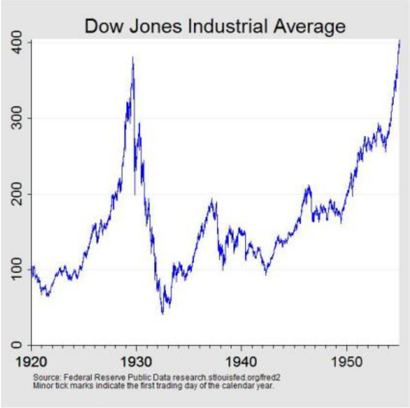
THREE U. S. C. STUDENTS ARE CRASH VICTIMS

CHRISTIANITY MUST DESTROY WAR OR WAR WILL OVERTHROW CHURCH, SAYS PASTOR SCHROCK

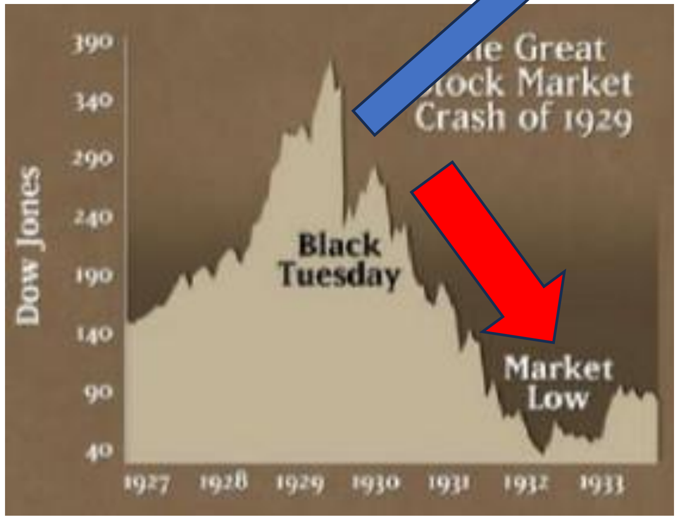


Irwings dept-deflation model (1933) for the great Depression





1 day of Self Amplifying panic

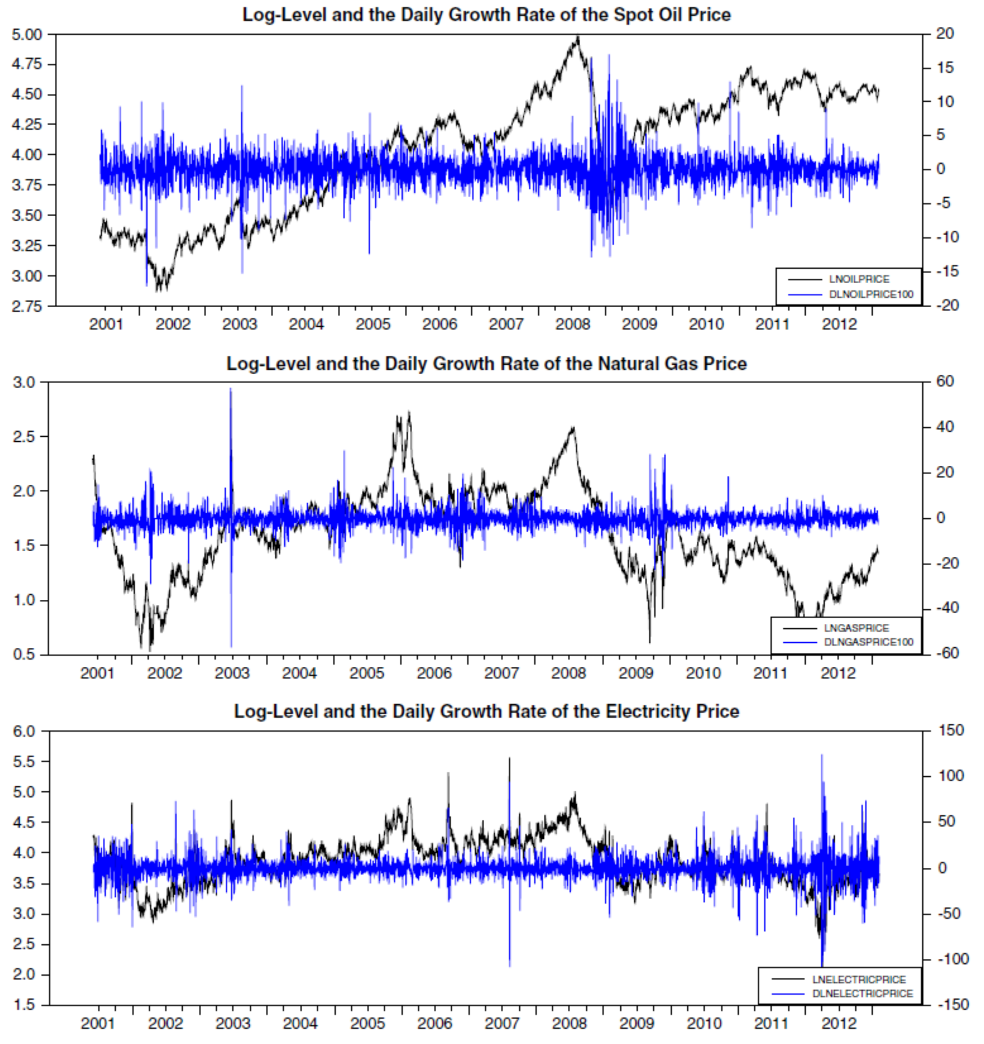


20% defaulted debt,
75% increase in $ value,
→150% increase in debt burden
From 1929 to 1933

… You cannot wait it out → disaster

universal but random features of stock markets:

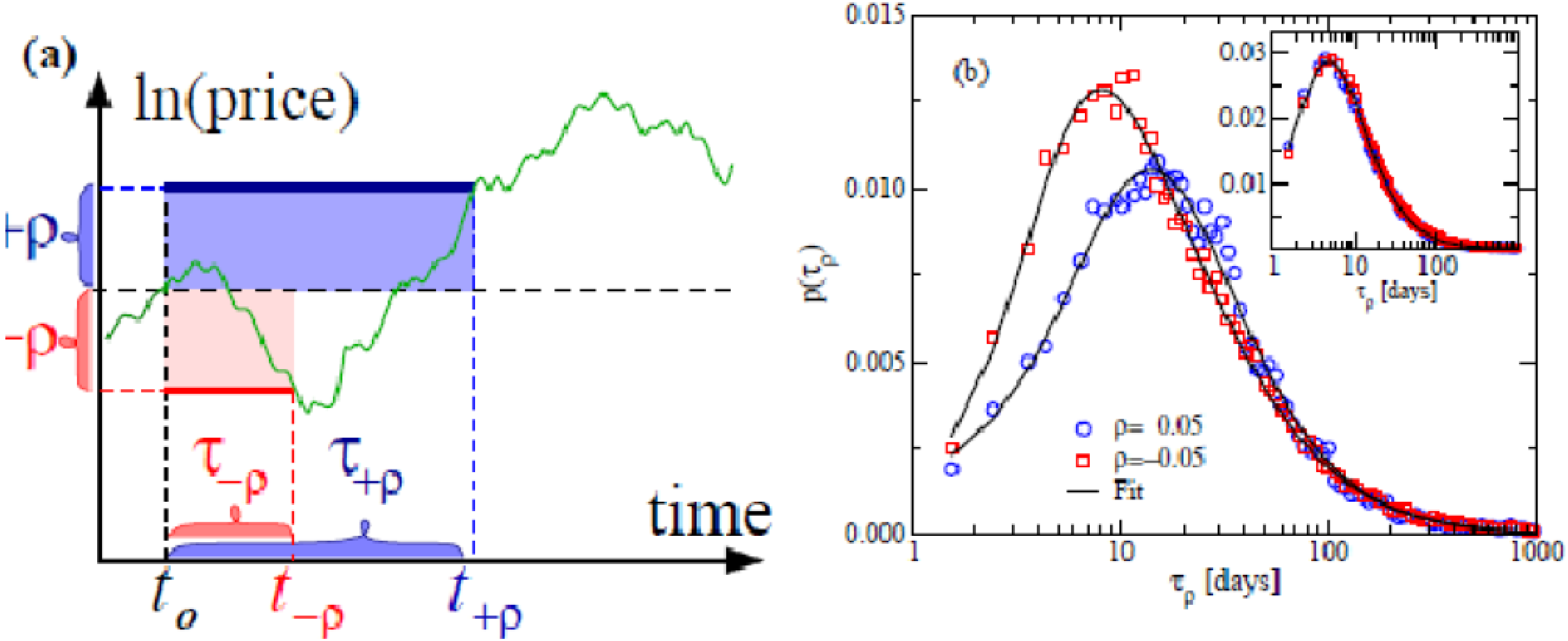
Correlation in size of change, not on its sign: The GARCH phenomena:



Variations is clustered,
Periods with big variation=
Big step size in random walk.
… that matter for bet-hedging
as we will see later.

Fear Factor model: … look at strike price

i.e. investment horizon until certain change is seen:

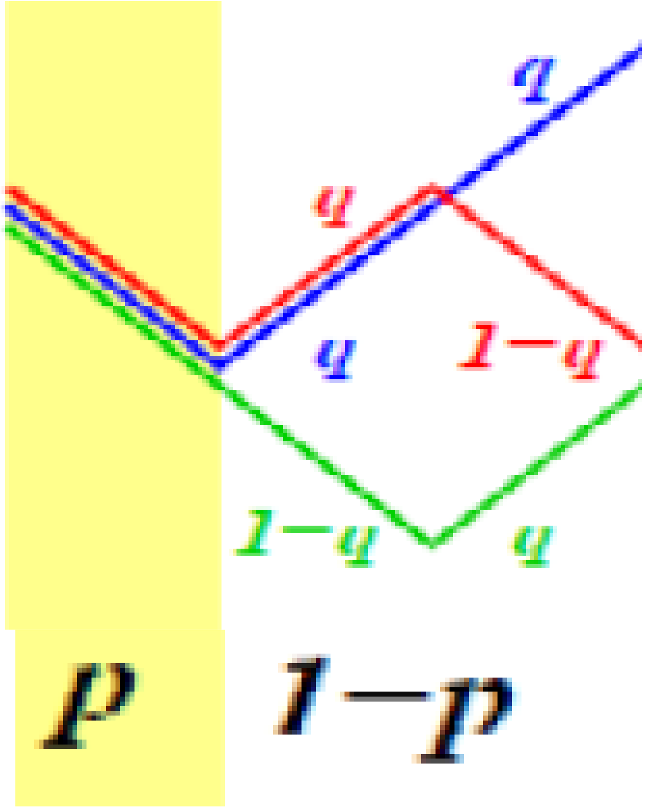


Tutorial

- Individual stocks symmetric’
- Sum typically reach -5% faster than it reach +5%

- How can that be?

Fear Factor model: Two parameters p,q



Reduced to 1 parameter by requirement of balanced random walk

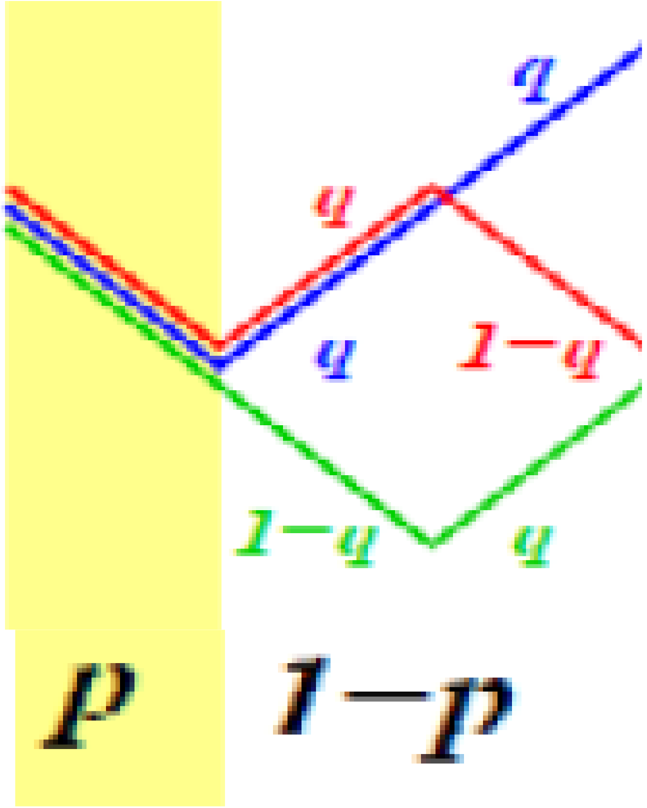
$$(1-p)\cdot q = p + (1-p)\cdot(1-q)$$

Probability
To go
Up
Require good
times (1-p)
And luck q

Can go
down
If panick
Event p

Can go
down
Even if good
Time (1-p)
Provided lack of luck
(1-q)

Fear Factor model: Two parameters p,q



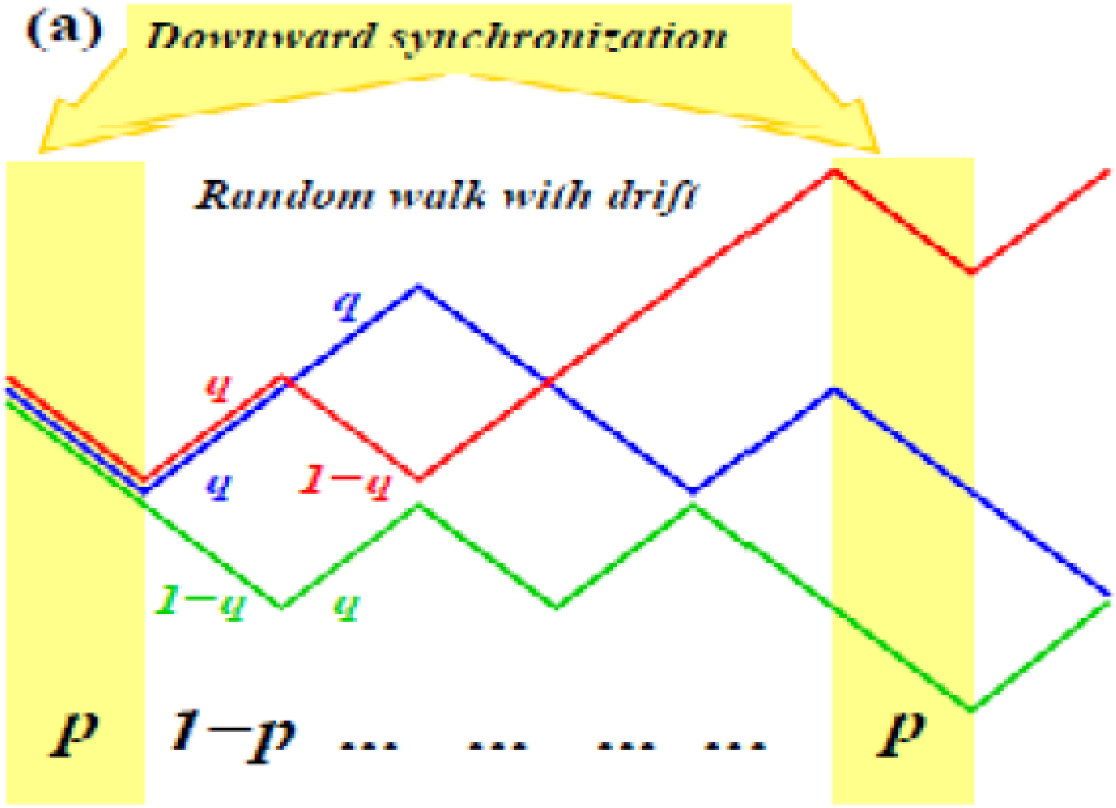
Reduced to 1 parameter by requirement of balanced random walk

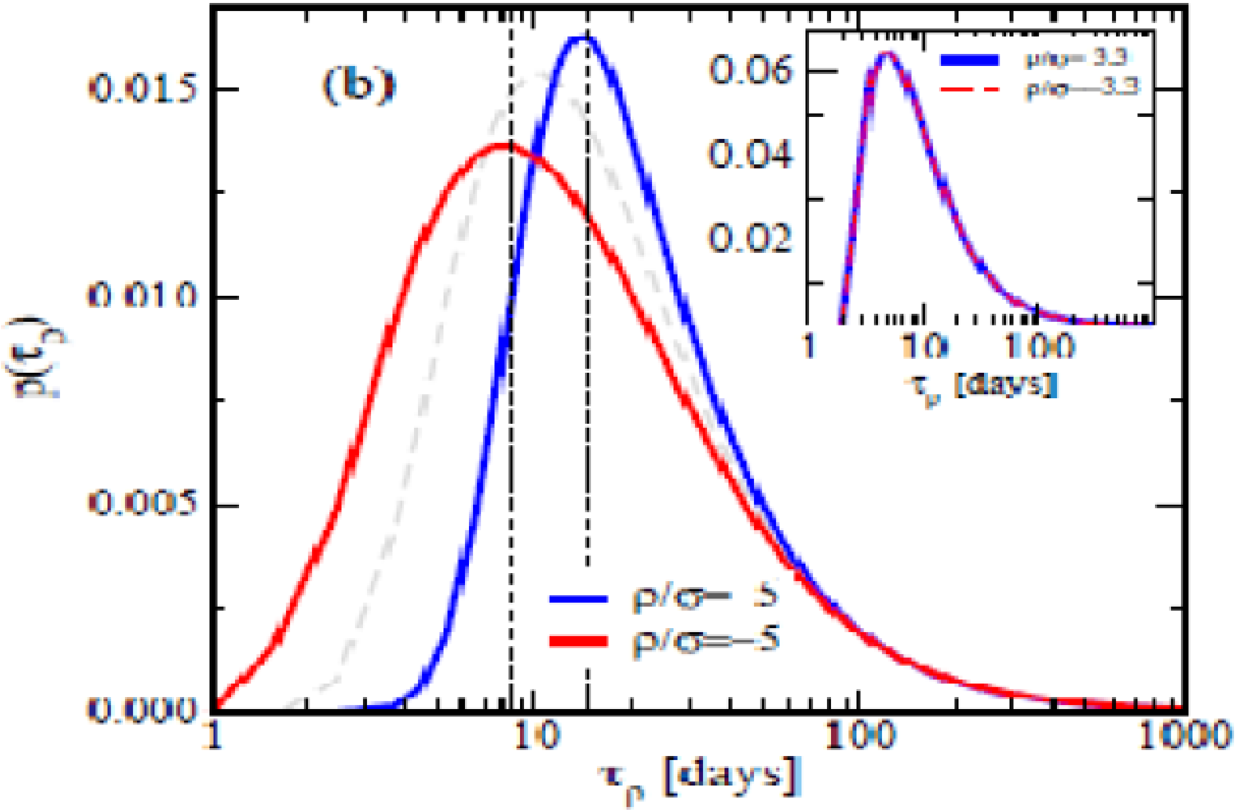
$$(1-p)\cdot q = p + (1-p)\cdot(1-q)$$

$$\Rightarrow \quad q = \frac{1}{2\cdot(1-p)}$$

i.e. if p is above ½ it get impossible to balance the bad times!

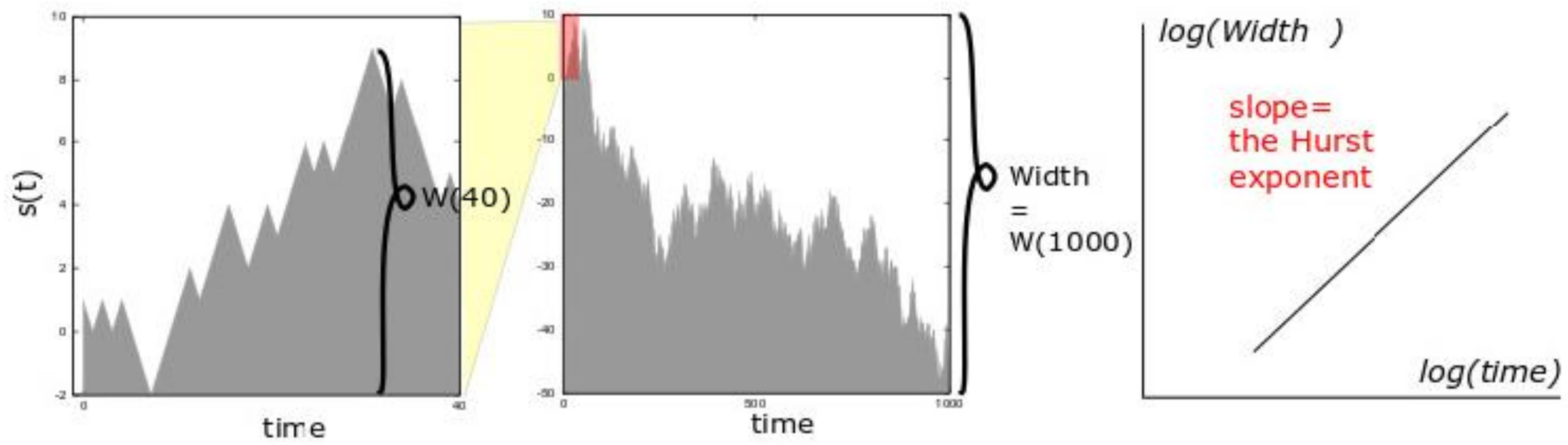
Model adjusted to fit peaks at 7 and 14 days respectively:





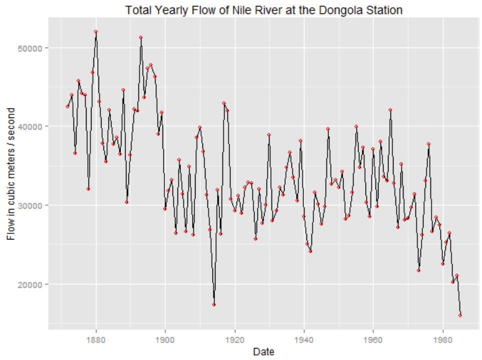
Summary **(from Monday)**

The "*Hurst*" exponent H (and walks with D=2-H)



$$\langle(\Delta s(T))^2\rangle = \langle(s(t+T)-s(t))^2\rangle_t \propto T^{2H}$$

Invented by Hurst studying water level in Nile:

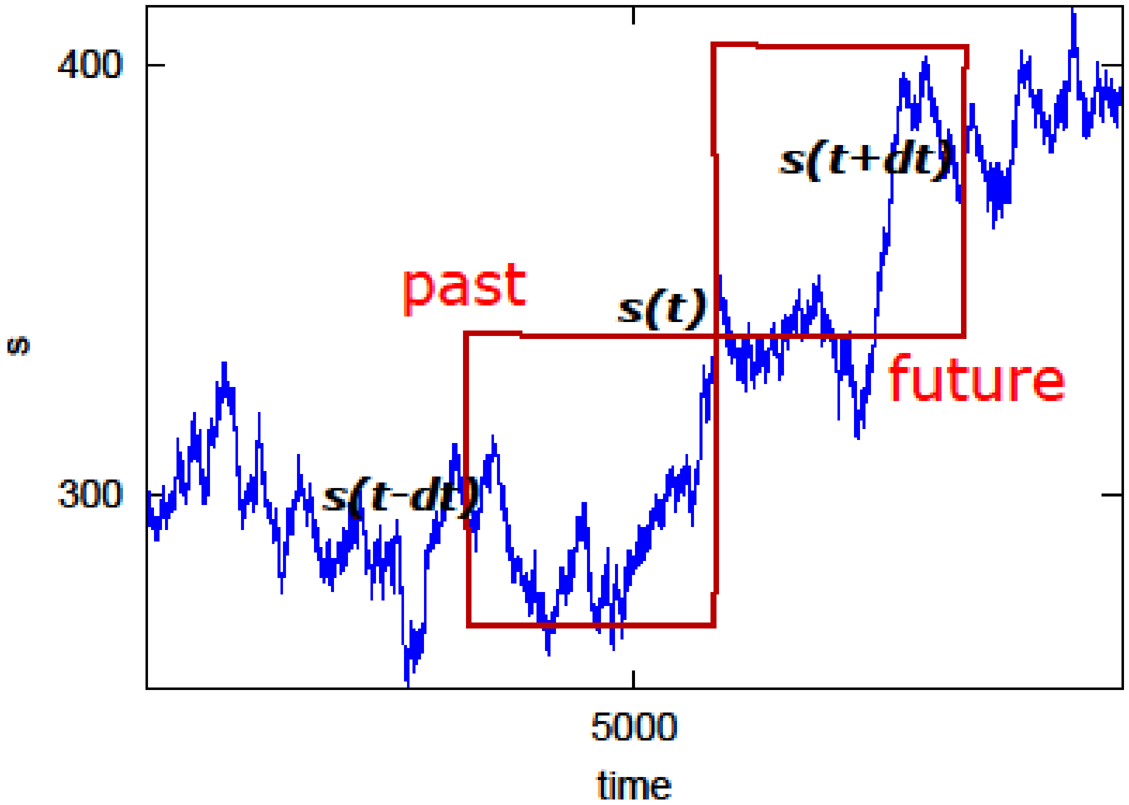


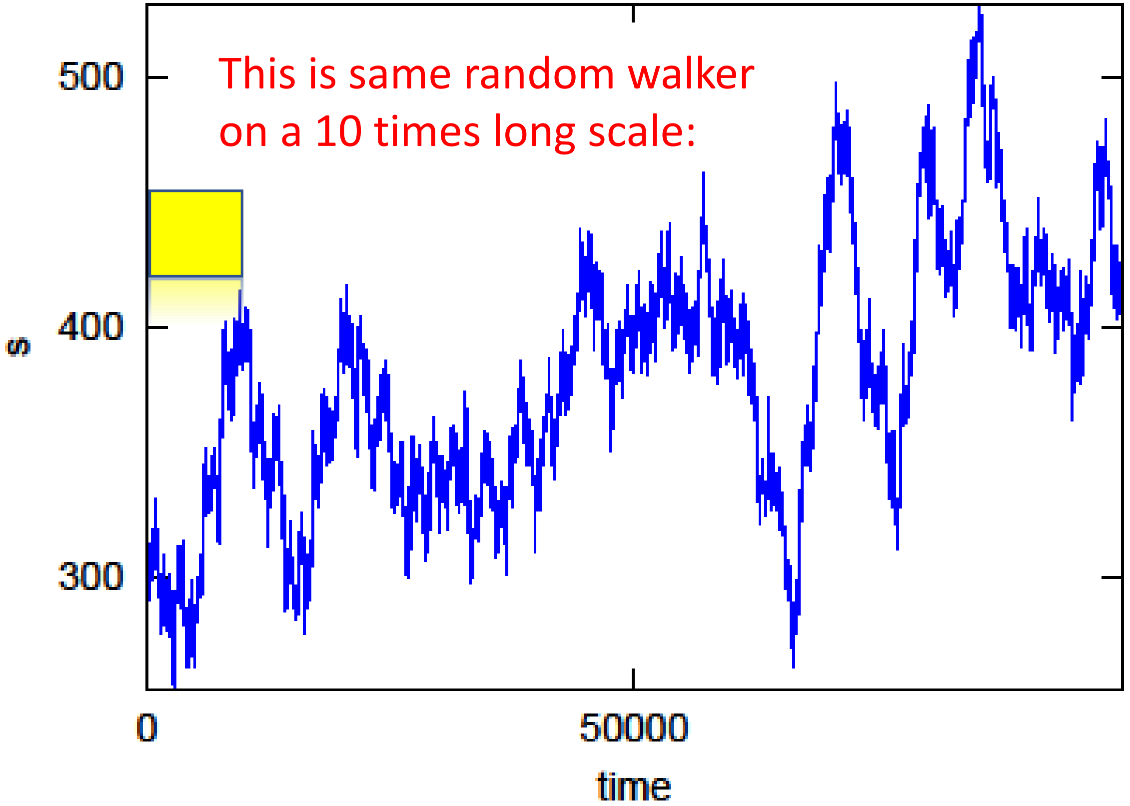
Here H = 0.70→0.75

Dimension: D=2-H

Summary **(Monday)** **:** Correlation between past and future:

$$C \equiv \frac{\langle (s(t) - s(t-T)) \cdot (s(t+T) - s(t)) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t} = \frac{\langle -\Delta s(-T) \cdot \Delta s(T) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t}$$





Summary (Monday); Correlation between past and future

$$
\begin{aligned}
f(T) &= \langle (\Delta s(T))^2 \rangle_x = \langle s^2(x+T) + s^2(x) - 2s(x)s(x+T) \rangle_x \\
&= 2(\langle s^2(x) \rangle - \langle s(x)s(x+T) \rangle_x) \propto T^{2H} ,
\end{aligned}
$$

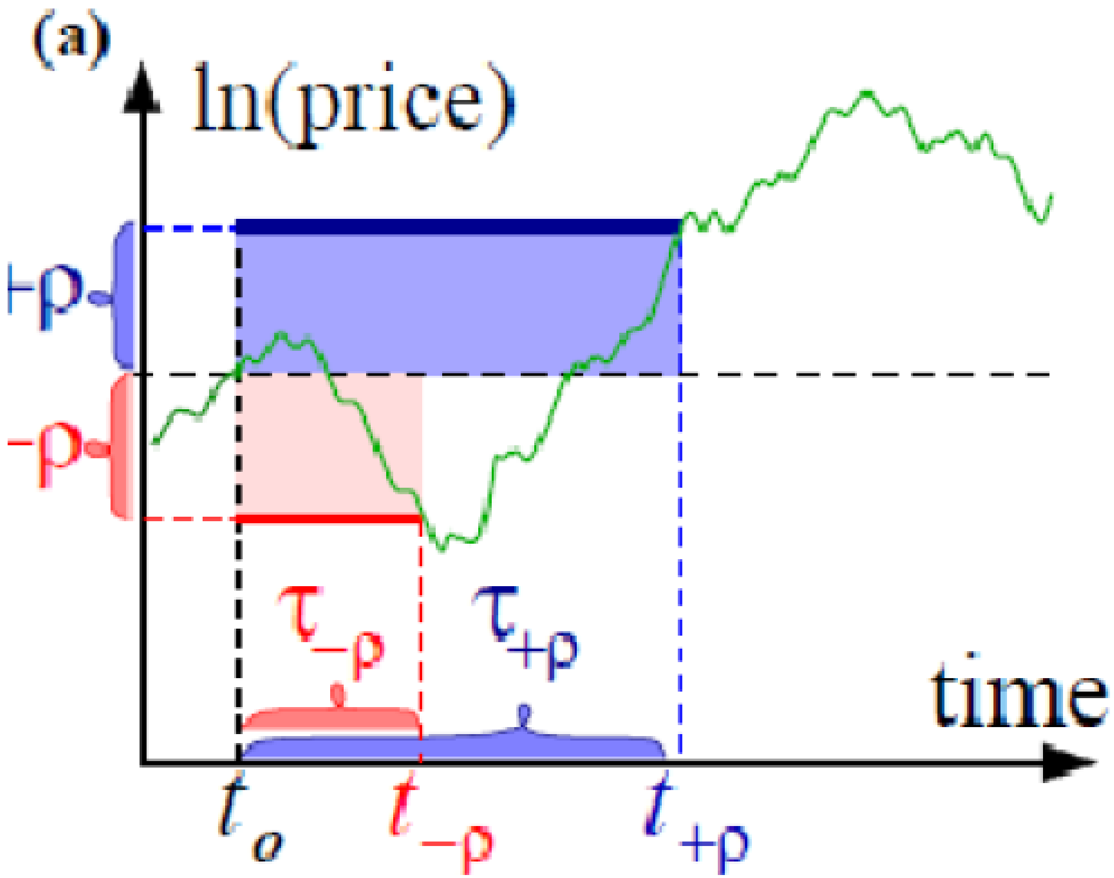
$$
C \equiv \frac{\langle (s(t) - s(t-T)) \cdot (s(t+T) - s(t)) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t} = \frac{\langle -\Delta s(-T) \cdot \Delta s(T) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t}
$$

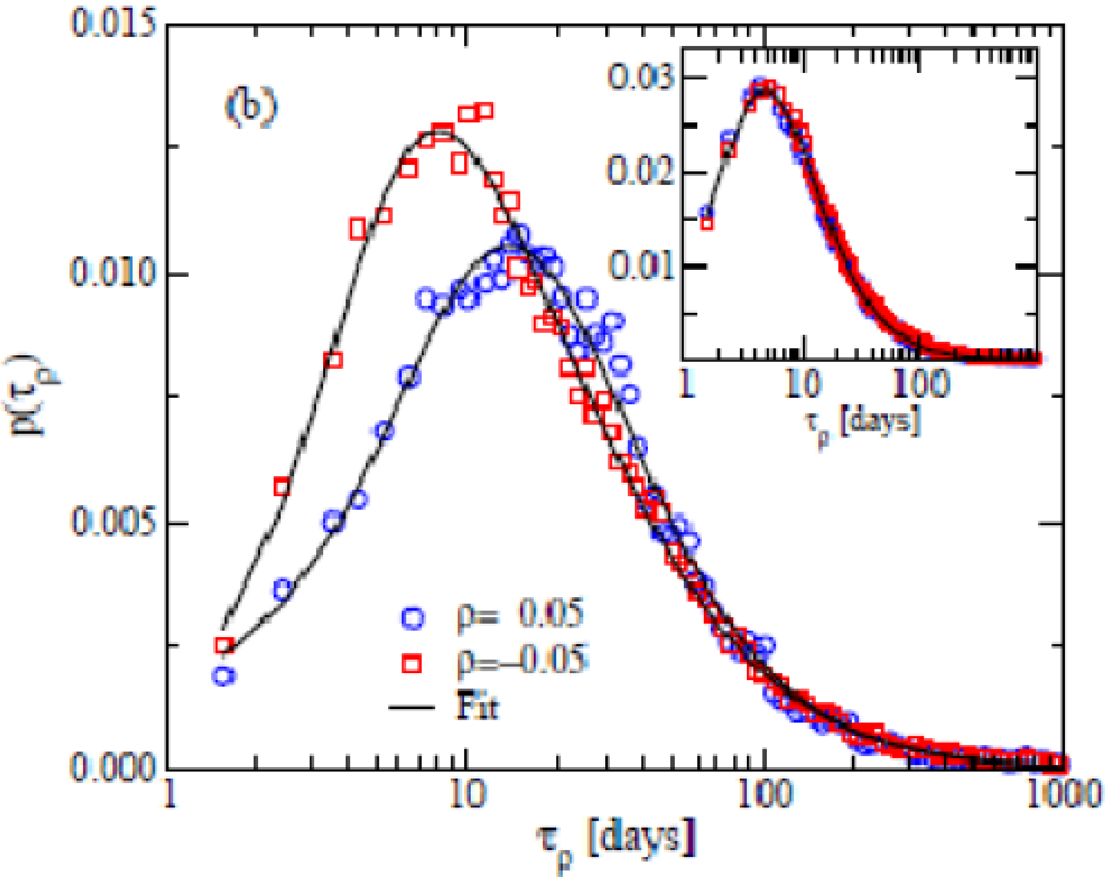
$$
f(T) = const \cdot T^{2H} \Rightarrow C = \frac{\langle -\Delta s(-T) \cdot \Delta s(T) \rangle_t}{\langle (\Delta s(T))^2 \rangle_t} = 2^{2H-1} - 1
$$

$$
H>0.5 \longrightarrow p_{\text{follow trend}} = 4^{H-1} > 0.5
$$

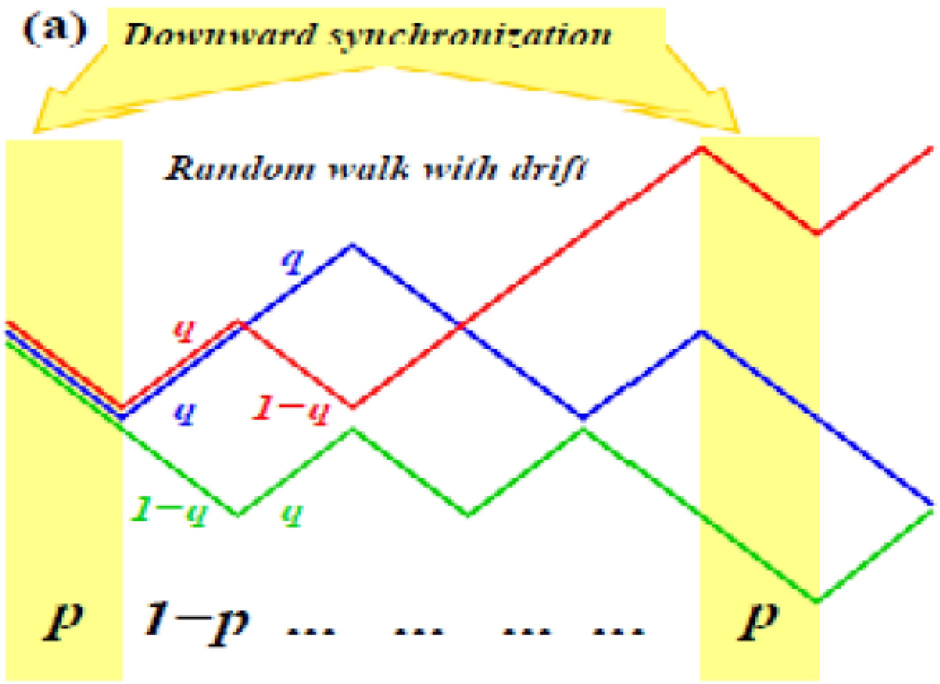
*Profit(betting on trend) = Pfollow-(1-Pfollow) = 2 Pfollow-1 = 2**(2H-1)-1 = C*

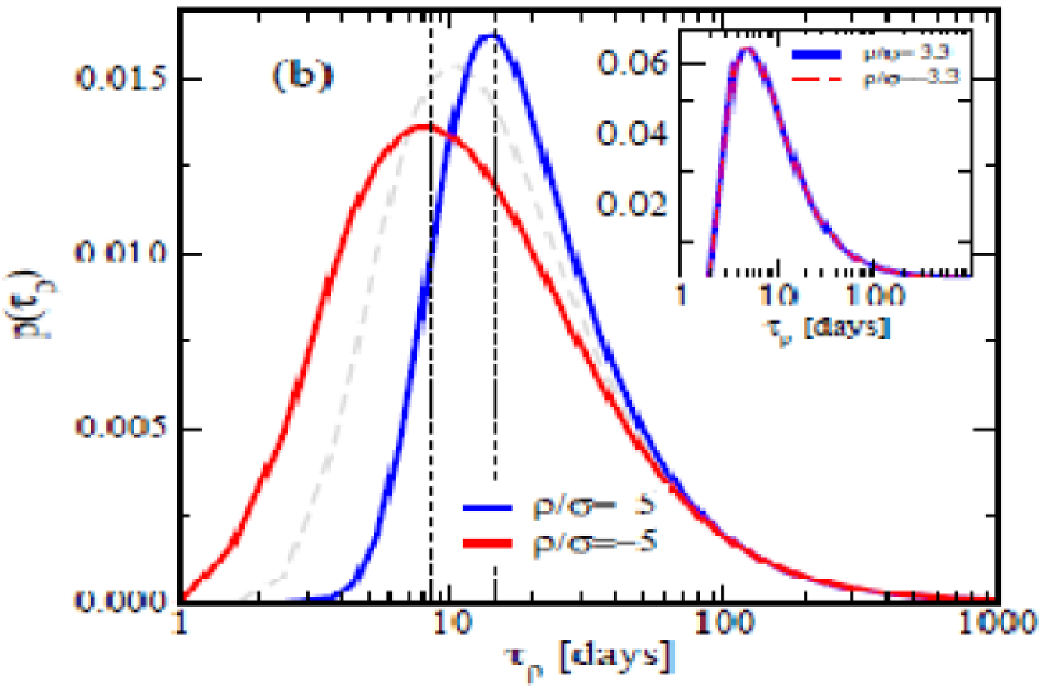
Summary (Monday): Collective fear revealed from statistics of first return prices





Summary: fear factor Model adjusted to fit peaks at 7 and 14 days





$$q = \frac{1}{2 \cdot (1 - p)}$$

Bet Hedging

- … We don’t know the future,….
- … So we have to bet strategically,….

Bet Hedging in markets with small changes

$$dv = \mu \cdot v \cdot dt + \sigma \cdot v \cdot dw$$

$$dw = w(t + dt) - w(t)$$

Into the future!

$$\langle (w(t) - w(t'))^2 \rangle = |t - t'|$$

And average w(t)-w(t')=0

$$s(t) = \log(v(t))$$

Average over all realizations of noise w (that in itself is modeled as a random walk)

$$
\begin{aligned}
ds &= \log v(t+dt) - \log v(t) \\
&= \log(v+dv) - \log(v) \\
&= \frac{1}{v}\cdot dv - \frac{1}{2v^2}(dv)^2 \\
&= \frac{1}{v}(\mu\cdot v\cdot dt + \sigma v\cdot dw) - \frac{1}{2v^2}(\sigma\cdot v\cdot dw)^2 \\
&= \left(\frac{1}{v}\cdot\mu\cdot v - \frac{1}{2v^2}\cdot\sigma^2\cdot v^2\right)\cdot dt + \frac{1}{v}\cdot\sigma\cdot v\cdot dw \\
&= \left(\mu - \frac{\sigma^2}{2}\right)\cdot dt + \sigma\cdot dw
\end{aligned}
$$

$dv = \mu\cdot v\cdot dt + \sigma\cdot v\cdot dw$

Only keep terms up to order dt or dw^2

$<dw^2> = dt$

$$s(t) - s(0) = (\mu - \frac{\sigma^2}{2})t + \sigma\int_0^t dw$$

$\langle (w(t)-w(t'))^2\rangle = |t-t'|$

Brownian walker which only grows as $\sqrt{t}$.

Can be ignored in long time limit

Therefore one get the typical long time return:

$$s(t) = s(0) + (\mu - \sigma^2/2) \cdot t$$

loss due to fluctuation

May become negative even if average $\mu > 0$

Risk cost on your average performance because 50% up and 50% down amounts to 1.5*0.5<1
…this is captured by the contribution of the second order term in the expansion of (log(v))

Notice the ``funny dimensions" YOU SUBTRACT THE AVERAGE AND THE VARIANCE NOT THE SPREAD!!!

NB: You can recover the average behavior

In terms of a Brownian walker B(t)= $\int_0^t dw$

Average over all
end point of Browian walker

$$\begin{aligned}\langle v(t)-v(t_0)\rangle &= exp((\mu-\frac{\sigma^2}{2})t)\cdot\int_B e^{\sigma B(t)}\cdot\left(\frac{1}{\sqrt{2\pi t}}e^{-B^2(t)/2t}\right)\cdot dB \\ &= exp((\mu-\frac{\sigma^2}{2})t)\cdot\frac{1}{\sqrt{2\pi t}}\int_B e^{-(B-\sigma t)^2/(2t)+\sigma^2 t/2}\cdot dB\end{aligned}$$

where $B(t) = \int_0^t dw(t')$ is the end point of a Brownnian walk starting at position $x=0$. Thus the integral above (blue) averages out to $exp(\sigma^2 t/2)$ and

Bet hedging, keep a fraction x in your pocket:

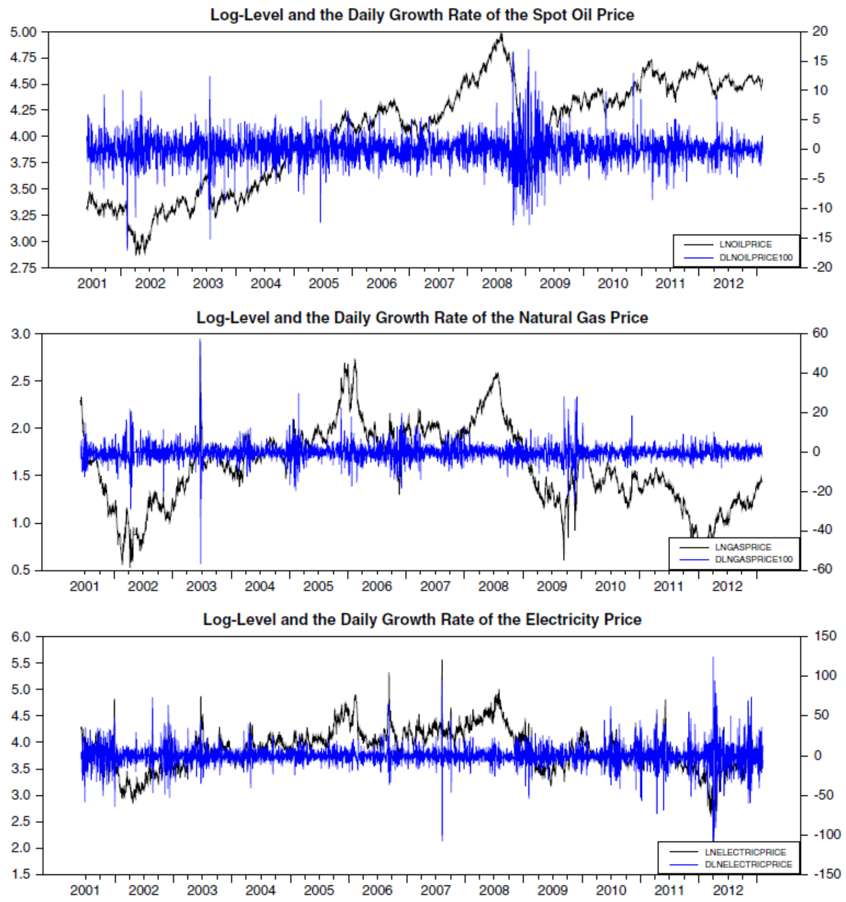
$$growth\ rate = \mu \cdot (1-x) - \frac{\sigma^2}{2} \cdot (1-x)^2$$

Optimal investment fraction: $1 - x = \frac{\mu}{\sigma^2}$

Only true when: $\langle (w(t) - w(t'))^2 \rangle = |t - t'|$

………………… thus variation should be smoothly varying with time!!

Fluctuations means something for investments,
…. Remember that fluctuations have structure

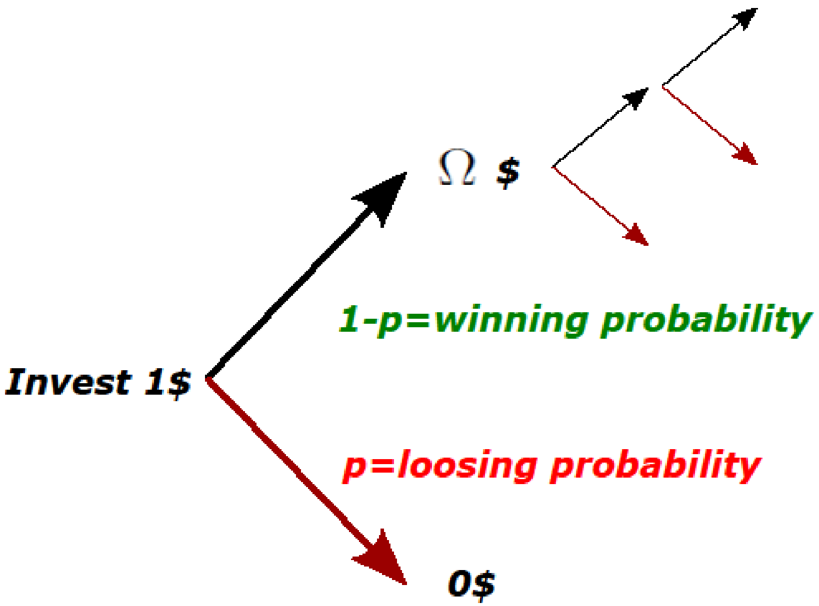


Mini Question:

What would 1000$
equivalent invested in a portfolio
of stocks at year 0AC in Rome be worth
today?

Mini Question: quit or quadruple

Consider game below for p=1/2 for loosing and potential gain =Omega =4:



You are given 100$, and infinite rounds of the game. How to play it optimally?

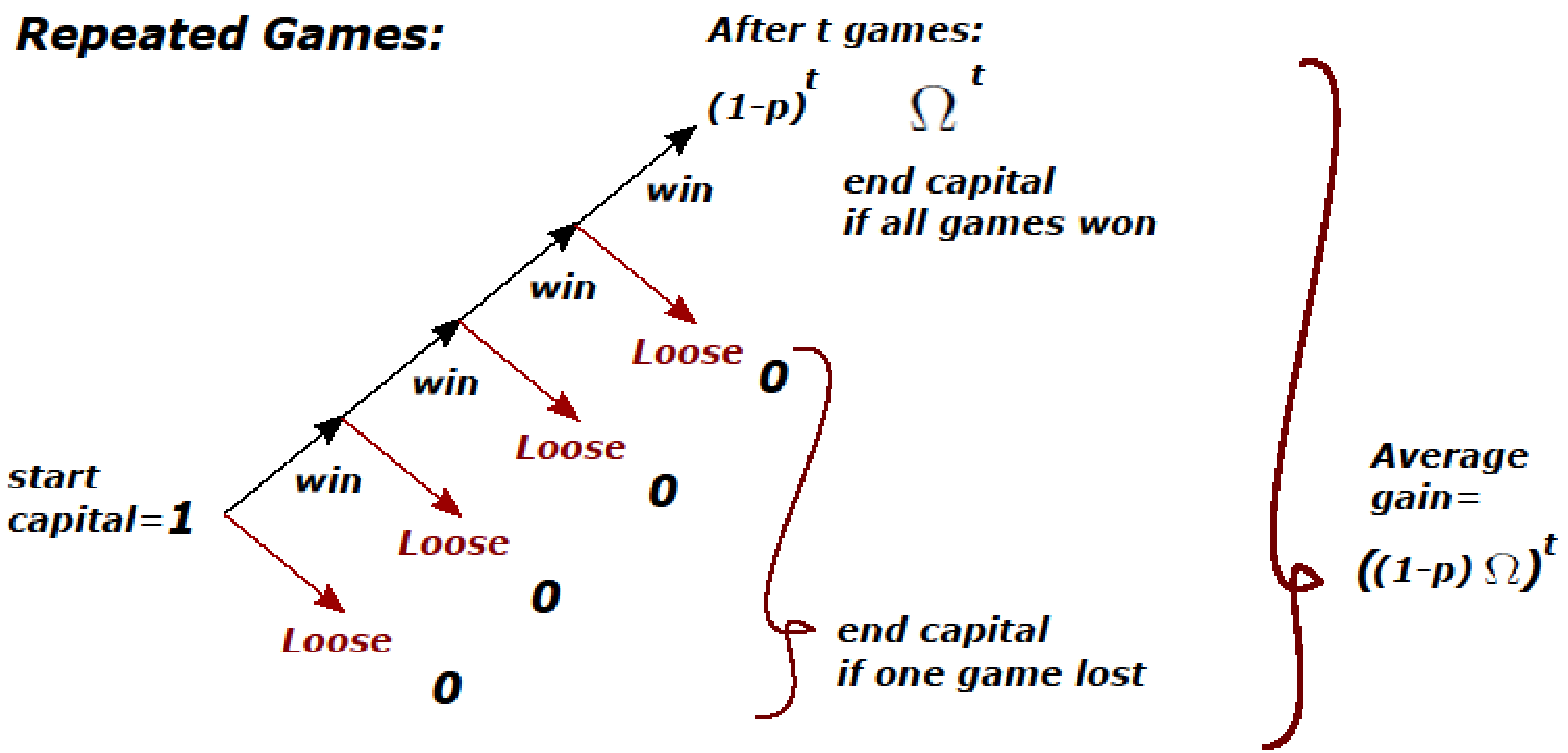
Average return

$$(1-p)\cdot\Omega + p\cdot 0 = (1-p)\cdot\Omega$$

Can easily be positive, and thus fortune grow as $2^{(1-p)\,\Omega\, t}$

But is that what is going to happen if you invest accordingly
Many-many times?

Investing everything:



$$propability(solvent|t) = (1-p)^t \rightarrow 0$$

But you have a fortune that exceeds your wildest dreams

P=1/2 analysis:

- After 2 games then 1$ typically will become:

$$Capital \propto Win \cdot Loose$$

- Put now x in bank and play with fraction 1-x:

$$Capital_{typical}(per\ 2\ games) \propto (\Omega(1-x)+x) \cdot x$$

If game is won

If game is lost

- Optimal capital gained if:

$$x = (1/2) \cdot \Omega/(\Omega - 1)$$

Tutorial:

Ω=1.5: x=…

Ω=2: x=1

Ω=3: x=…?

Ω=100: x=?

$$x = (1/2) \cdot \Omega/(\Omega - 1)$$

Generalizing the equation:

$$\langle Cap(t) \rangle = Cap(0) \cdot \sum_{b=0}^{t} \left(\frac{t!}{(t-b)!b!} \right) \cdot p^b \cdot (1-p)^{t-b} ((1-x)\Omega + x)^{t-b} x^b$$

Average
capital
After
t games

Chance to loss
b times out of *t*

Gain factor if loss
b times out of *t* (and win *t*-*b* times)
…. If x=0 this only becomes >0 for b=0

Red part defines typical

This has an expected
p ∗ t bad events

$$Cap_{typical}(t) \quad \propto \quad ((1-x)\Omega + x)^{t-pt} x^{pt}$$

Capital gain:

$$
\begin{aligned}
Cap_{typical}(t) &\propto ((1-x)\Omega + x)^{t-pt} x^{pt} \\
&= \left(win^{1-p} \cdot loose^{p}\right)^{t} \\
&= \left(((1-x)\Omega + x)^{1-p} x^{p}\right)^{t} = e^{t\Lambda(x)}
\end{aligned}
$$

With growth rate:

$$\Lambda(x) = (1-p)\cdot\log(\Omega(1-x)+x) + p\cdot\log(x)$$

Generalization from p=1/2 before

$$
\begin{aligned}
\frac{d\Lambda(x)}{dx}\Big|_{x^*} &= -(1-p)\cdot\frac{\Omega-1}{\Omega(1-x^*)+x^*} + p\cdot\frac{1}{x^*} = 0 \Rightarrow \\
x^* &= p\cdot\frac{\Omega}{\Omega-1}
\end{aligned}
$$

The larger p, the less you gamble, i.e. (1-x) smaller
…never gamble more than p

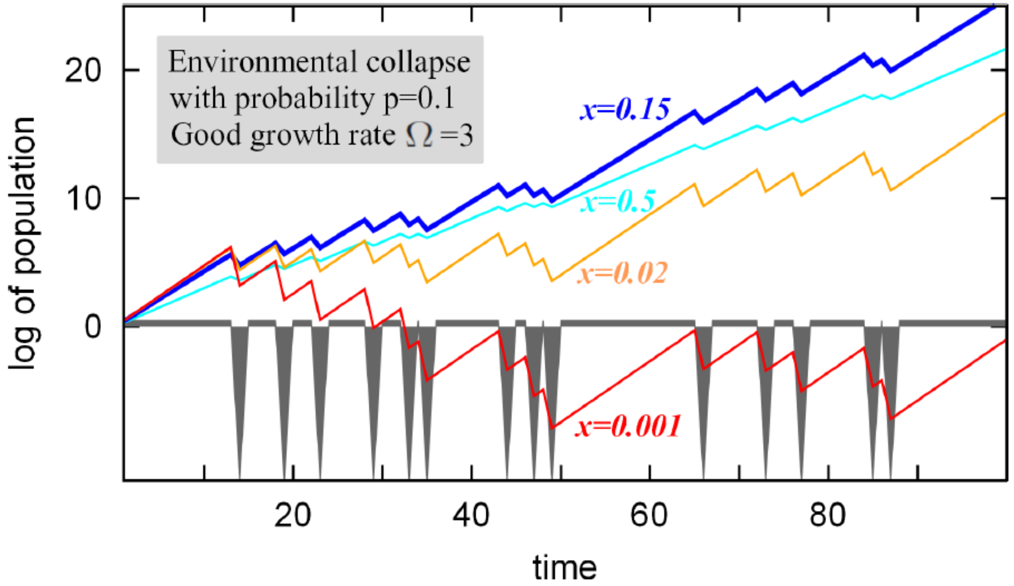
Simulating the betting timeseries:

- You chose a betting fraction 1-x (safe fraction x)
- At time 0 you have capital C(0)=1
- At each timestep the computer generate a loss (p) or win (1-p):

  if loss your fortune is adjusted to $C(t)=C(t-1)*x$

  if win your fortune is adjusted to $C(t)=C(t-1)*[(1-x)\mathbf{\Omega}+x]$

Plot C(t) versus t for different realizations:



Generalize the equation:

$$\langle Cap(t)\rangle = Cap(0)\cdot\sum_{b=0}^{t}\left(\frac{t!}{(t-b)!b!}\right)\cdot p^b\cdot(1-p)^{t-b}((1-x)\Omega+x)^{t-b}x^b$$

to the case where loss means
one retain a fraction w of gambled fortune

What is then:

$$x^* = p\cdot\frac{\Omega}{\Omega-1} \Rightarrow ?$$

“Funny” equation when Omega>>1:

Bet hedge fraction= P(loss) - fraction left if one lose

$$x=p-w$$

Summary of last lectures:

- Markets makes close to random walk
- Hurst exponent H=1/2 means that there is no obvious correlations
- Down movements more correlated than up movements

$$
\begin{aligned}
ds &= \log v(t+dt) - \log v(t) \\
&= \log(v+dv) - \log(v) \\
&= \frac{1}{v}\cdot dv - \frac{1}{2v^2}(dv)^2 \\
&= \frac{1}{v}(\mu\cdot v\cdot dt + \sigma v\cdot dw) - \frac{1}{2v^2}(\sigma\cdot v\cdot dw)^2
\end{aligned}
$$

$$dv = \mu\cdot v\cdot dt + \sigma\cdot v\cdot dw$$

Only keep terms up to order dt or dw^2

$$< dw^2 > = dt$$

$$s(t) - s(0) = (\mu - \frac{\sigma^2}{2})t + \sigma\int_0^t dw$$

$$\langle (w(t) - w(t'))^2\rangle = |t - t'|$$

Brownian walker which only grows as $\sqrt{t}$.

Can be ignored in long time limit

Therefore one get the typical long time return:

$$s(t) = s(0) + (\mu - \sigma^2/2) \cdot t$$

loss due to fluctuation

May become negative even if average $\mu > 0$

Risk cost on your average performance because 50% up and 50% down amounts to 1.5*0.5<1
…this is captured by the contribution of the second order term in the expansion of (log(v))

Notice the ``funny dimensions" YOU SUBTRACT THE AVERAGE AND THE VARIANCE NOT THE SPREAD!!!

Extreme Games (Good><bad):

Equation when Omega>>1:

Bet hedge fraction= P(loss) - fraction left if one lose

$$x=p-w$$

Topics

- 1: Equilibrium and detailed balance: T=1/(dS/dE) & H=...→ and Z, F, <E> and S,

  Ergodicity and P(a→b)*p(a)=P(b→a)*p(b)→ Metropolis sampling with aim to calculate Z and <E> and...
- 2: Ising model, phase transition, critical point, mean field model>< 1-d model.
- 3: Scaling, percolation, critical exponents (P(s)=1/s^2.5), fractals, begin to introduce time (directed percolation)
- 4: Random walks, Critical branching & Self organized criticality, **Separated timescale dynamics.**
- 5: Networks & critical spreading on networks & scale free networks & pattern in networks & construction and emergence of scale free networks, **Limited and nearly fixed interactions, heterogeneity.**
- 6: Stochastic simulations and Agent based models. → **……d/dt physics in discrete space:**

  Non equilibrium ising like model & epidemic model, **quenched disorder, heterogeneity.**
- 7: Econophysics: random walks in log space, Hurst exponents, aspects of bet hedging. Models to think about stock price fluctuations.

→ **From equilibrium to dynamics**

→ **…a lot of randomness… type matter: *dx/dt=noise(t)* or *dx/dt=noise(x)* (the latter quenched and better dealt with in agent based models / networks)**

- **1) Stat. Mechanics**,Entropy, Partition function, Ising model, and Metropolis Simulations (ch 1).
- **2) Phase transitions & Critical phenomena:** Mean Field Solution to Ising mode, Mean Field Potts model, 1-d Ising model and transfer matrix method (ch 2).
- **3) Percolation and fractals:** Percolation one a Bethe lattice with some critical exponents, renormalization, fractals, equations for fractal dimensions (ch. 3)
- **4) Self-organized criticality:** First return of Random walk, critical branching, the sandpile model, the evolution model (ch. 4)
- **5) Networks:** Amplification factor, scale-free networks, analyzing patterns of networks, algorithms to make or evolve scale-free networks (ch. 5)
- **6) Agent-based models & Event-based simulation:** simulation with discrete but random changes, Gillespie algorithm, Example of agent-based simulation and its advantages (ch. 6). SIR models and advantages of agent-based models in epidemic models. Excitable media is not persum.
- **7) Econophysics:** Hurst exponent, correlation between past and future, Fear Factor model, Bet hedging approaches (ch. 7)

Cross subject Themes:

- …. Connected *in part* by history (of invention 1924-1980 (Ising model)
  → 1980 (chapter 3) → 1990 (chapter4) → 2000-2020 (chapter 5 & 6 & 7))

- …. Connected *in part* by power laws

- …. *In part* by universality… one rule (ring) to rule them all…. Here saw some power laws that are recurrent, 1/x^2 and 1/x^1.5

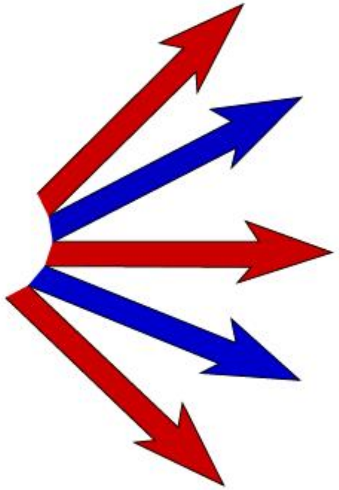
- … universal methodology from physics to diseases to social network to economic systems

- From equilibrium towards various forms of self organization (using time… lots of time)

Complex Physics (2024)
``More is Different”

More is different:

$$F \propto \frac{Q_a\, Q_b}{R^2}$$



….one force as theory of everything (living?)

Complex Physics (2024)
``More is Different”

…Stochastic dynamics with mixed time scales

…connecting phenomena at different length scales

Large systems=many degrees of freedom=more

Equilibrium → Non equilibrium

Organized by Energy → by Time & excitation

From disorder to some order=is different

…in form of far-reaching correlations=power laws/fractals

Cross subject themes: the recurrent return of random walks

- Cluster size in Bethe lattice (ch. 3), Critical branching+SOC (ch 4), Merging and Injection (ch. 5), Voter models & 1-d opinion models (Ch 6), Econophysics (Hurst exponent, strike price distribution, Ch. 7)
- Self organization: Time can organize patterns far easier than potentials (SOC (ch. 4), Networks far from equilibrium (ch. 5)), Positive feedback & tipping points (ch. 6), Equilibrium & non equlibrium markets (ch. 7), Bet hedging.
- Randomness may have structure, e.g. Noise(t) ←→ Noise(x). Example is superspreaders or whatever dynamics on fixed heterogeneous networks. (quenched dx/dt=noise(x), or annealed dx/dt=noise(t) very different)
- →next chapter (8)… Interfaces, random walks of random walks

Overall: ``More is different”

- Non equilibrium physics > > equilibrium physics
- Complexity science …. like non-elephant zoology
- … Many topics should inspire you to think about the world in terms of physics models, and also think about the social and biological systems
- How you think about dynamics and noise matters, … separated timescales and quenched noise matters….
- Life/dynamics is qualitatively & and quantitatively much more than mean field models